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Overview of functions



2/2 SINUMERIK 828D and SINUMERIK 828D Smart CNCs

2/2 Control structure and configuration
2/5 Connectable drives
2/6 Connectable measuring systems
2/7 Connectable motors
2/8 Axis functions
2/9 Spindle functions
2/10 Interpolations
2/13 Couplings
2/14 Transformations
2/15 Measuring functions and measuring cycles
2/15 Technologies
2/18 Motion-synchronous actions
2/19 Engineering
2/20 CNC programming language
2/23 Programming support
2/26 Simulations
2/27 Operating modes
2/29 Tools
2/30 Tool management
2/31 Communication and data management
2/32 Operation
2/35 Monitoring functions
2/37 Compensations
2/38 PLC area
2/40 Safety functions
2/41 Commissioning
2/42 Diagnostic functions
2/42 Remote control of other systems
2/42 Service and maintenance
2/42 SINUMERIK Ctrl-Energy
2/43 Overview of CNC options
2/43 Manufacturer options
2/50 User options
2/54 System overview

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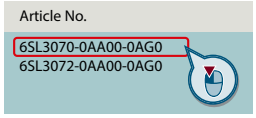
The functionality of SINUMERIK 828D CNCs complies with the export list restrictions. Accordingly, these CNCs do not require official approval in accordance with EU or German law as a result of their type.

The following overview lists all the functions which are available with SINUMERIK 828D CNCs.

The information in the overview of SINUMERIK 828D functions is based on the following software version:

CNC	Software version
SINUMERIK 828D	5.25 SP3

Click on an Article No. in the catalog PDF to call it up in SiePortal and to obtain all the information.



Or directly on the internet, e.g.
www.siemens.com/product_catalog_DIMC?6SL3070-0AA00-0AG0

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Control structure and configuration

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Control structure and configuration								
Panel-based compact CNC comprising:								
• Compact operator panel CNC		✓	✓	–	✓	–	✓	–
• CNC/PLC control unit		✓	✓	✓	✓	✓	✓	✓
• Closed-loop control for drives		5	5	5	6	6	6	6
Design, drive-based/PC-based		–	–	–	–	–	–	–
Operator panel CNC:								
• Horizontal		✓	✓	–	✓	–	✓	–
• Vertical		✓	✓	–	✓	–	✓	–
• Color display		12.1"	12.1"/ 15.6"	–	12.1"/ 15.6"	–	12.1"/ 15.6"	–
• Integrated QWERTY keyboard with short-stroke keys		✓	✓	–	✓	–	✓	–
SINUMERIK operator panels with TCU		–	–	–	–	–	–	–
SINUMERIK operator panels with PCU		–	–	✓	–	✓	–	✓
SINUMERIK 828D:								
• PPU 260.5	**6FC5370-3AA50-0AA0**	O	–	–	–	–	–	–
• PPU 261.5	**6FC5370-2AA50-0AA0**	O	–	–	–	–	–	–
• PU 272.5	**6FC5370-7AA50-0AA0**	–	–	O	–	O	–	O
• PPU 270.5	**6FC5370-6AA50-0AA0**	–	O	–	O	–	O	–
• PPU 271.5	**6FC5370-5AA50-0AA0**	–	O	–	O	–	O	–
• PPU 290.5	**6FC5370-8AA50-0BA0**	–	O	–	O	–	O	–
System software, export version, on SD card, with license								
• SINUMERIK 828D with PPU 260.5/PPU 261.5/PPU 270.5/PPU 271.5/ PPU 290.5/PU 272.5:								
- Turning	6FC5835-1GY50-0YA0	O	O	O	–	–	–	–
- Milling	6FC5835-2GY50-0YA0	O	O	O	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface	6FC5835-3GY50-0YA0	O	O	O	–	–	–	–
- Turning	6FC5834-1GY50-0YA0	–	–	–	O	O	–	–
- Milling	6FC5834-2GY50-0YA0	–	–	–	O	O	–	–
- G-Tech Cylindrical/G-Tech Surface	6FC5834-3GY50-0YA0	–	–	–	O	O	–	–
- Turning	6FC5836-1GY50-0YA0	–	–	–	–	–	O	O
- Milling	6FC5836-2GY50-0YA0	–	–	–	–	–	O	O
- G-Tech Cylindrical/G-Tech Surface	6FC5836-3GY50-0YA0	–	–	–	–	–	O	O
Software for SIMATIC IPC SINUMERIK Operate		–	–	O	–	O	–	O
SINUMERIK Operate Embedded HMI		✓	✓	–	✓	–	✓	–
Windows-based HMI		–	–	✓	–	✓	–	✓
DRIVE-CLiQ drive interface		–	✓	✓	✓	✓	✓	✓
PROFINET drive interface (PLC periphery interface)		✓	✓	✓	✓	✓	✓	✓

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Control structure and configuration (continued)								
Numeric Control Extension NX10.3 for applications with up to 9 axes or for reducing the current controller cycle clock to 62.5 µs:	**6SL3040-1NC00-0AA0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Numeric Control Extension NX15.3 for applications with up to 12 axes or for reducing the current controller cycle clock to 62.5 µs:	**6SL3040-1NB00-0AA0**							
• Turning		–	–	–	O	O	O[1]	O[1]
• Milling		–	–	–	O	O	O[1]	O[1]
• G-Tech Cylindrical/G-Tech Surface		–	–	–	O	O	O[1]	O[1]
Machining channel, each additional	**6FC5800-0AC10-0YH0**	–	–	–	–	–	O	O
Machining channel, maximum:								
• Turning	Basic configuration: 1 machining channel	1	1	1	1	1	2	2
• Milling	Basic configuration: 1 machining channel	1	1	1	1	1	2	2
• G-Tech Cylindrical/G-Tech Surface	Basic configuration: 1 machining channel	1	1	1	1	1	2	2
Mode group, maximum:								
• Turning	Basic configuration: 1 mode group	1	2	2	2	2	4	4
• Milling	Basic configuration: 1 mode group	1	2	2	2	2	4	4
• G-Tech Cylindrical/G-Tech Surface	Basic configuration: 1 mode group	1	2	2	2	2	4	4
Mode group, each additional:	**6FC5800-0AC00-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
CNC user memory (buffered) for CNC part programs:								
• Turning		3 MB	3 MB	3 MB	5 MB	5 MB	10 MB	10 MB
• Milling		3 MB	3 MB	3 MB	5 MB	5 MB	10 MB	10 MB
• G-Tech Cylindrical/G-Tech Surface		3 MB	3 MB	3 MB	5 MB	5 MB	10 MB	10 MB
CNC user memory expanded for programs	**6FC5800-0AP77-0YH0**	–	–	–	O 100 MB	O 100 MB	O 100 MB	O 100 MB
Execution from external storage EES Only limited by the available memory with the following options: Execution from external storage EES (6FC5800-0AP75-0YH0) or expansion of the CNC user memory (6FC5800-0AP77-0YH0)	**6FC5800-0AP75-0YH0** Contains the option: CNC user memory expanded for programs 6FC5800-0AP77-0YH0	–	–	–	O	O	O	O

[1] When using 2 × Numeric Control Extensions NX15.3, applications with up to 18 axes are possible.

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Control structure and configuration

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Control structure and configuration (continued)								
HMI user memory additional 256 MB on SD card of the PPU		–	–	–	–	–	–	–
Additional CNC user memory 300 MB, extended Max. extension 1000 MB	**6FC5800-0AD60-0YH0** Requirement: Execution from external storage EES 6FC5800-0AP75-0YH0	–	–	–	O	O	O	O
Axes/spindles or positioning axis/auxiliary spindle:								
• Basic configuration axes/spindles:								
- Turning		3	3	3	3	3	3	3
- Milling		4	4	4	4	4	4	4
- G-Tech Cylindrical/G-Tech Surface		3	3	3	3	3	3	3
• Maximum axes in system (including axes/spindles in machining channel + axes in handling axes channel + PLC positioning axes):								
- Turning		5	8	8	11	11	18	18
- Milling		5	8	8	11	11	16	16
- G-Tech Cylindrical/G-Tech Surface		5	8	8	11	11	18	18
• Maximum number of axes/spindles + PLC positioning axis in machining channel(s):								
- Turning		5	5	5	6+2	6+2	10+2 [1]	10+2 [1]
- Milling		5	5	5	6+2	6+2	8+2 [2]	8+2 [2]
- G-Tech Cylindrical/G-Tech Surface		5	5	5	6+2	6+2	10+2 [1]	10+2 [1]
• Maximum number of axes in handling channel(s):		–	4	4	4	4	4+4	4+4
Axis/spindle, each additional	**6FC5800-0AC20-0YH0**	O	O	O	O	O	O	O
Positioning axis/auxiliary spindle, each additional	**6FC5800-0AC30-0YH0**	O	O	O	O	O	O	O
Handling channel, each additional	**6FC5800-0AC40-0YH0**	–	O	O	O	O	O	O
Handling channel, maximum:								
• Turning		–	1	1	1	1	2	2
• Milling		–	1	1	1	1	2	2
• G-Tech Cylindrical/G-Tech Surface		–	1	1	1	1	2	2
PLC-controlled axis		✓	✓	✓	✓	✓	✓	✓
PLC positioning axis, e.g. CU310-2/320-2 PN via I/O interface based on PROFINET, maximum		–	–	–	2	2	2	2
Systems and number of the TOOLCARRIER:								
• Turning		–	–	–	6	6	6	6
• Milling		1	1	1	1	1	2	2
• G-Tech Cylindrical/G-Tech Surface		2	2	2	2	2	4	4
CNC lock function	**6FC5800-0AP76-0YH0**	O	O	O	O	O	O	O

[1] Max. number of axes in machining channels: 10 + 2 × positioning axis/auxiliary spindle 6FC5800-0AC30-0YH0.

[2] Max. number of axes in machining channels: 8 + 2 × positioning axis/auxiliary spindle 6FC5800-0AC30-0YH0.

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Connectable drives								
SINAMICS S120 Motor Modules via DRIVE-CLiQ	See SiePortal	–	✓	✓	✓	✓	✓	✓
SINAMICS S120 DRIVE-CLiQ on motor	See Motion Control Encoder measuring systems	–	✓	✓	✓	✓	✓	✓
SINAMICS S200 PN on motor	See SINUMERIK 828D Smart system	✓	–	–	–	–	–	–
SINAMICS S120 Sensor Module Cabinet-Mounted:	See SiePortal							
• SMC20	**6SL3055-0AA00-5BA3**	–	O	O	O	O	O	O
• SMC30	**6SL3055-0AA00-5CA2**	–	O	O	O	O	O	O
• SMC40	**6SL3055-0AA00-5DA0**	–	O	O	O	O	O	O
SINAMICS S120 Sensor Module External:	See SiePortal							
• SME20	**6SL3055-0AA00-5EA3**	–	O	O	O	O	O	O
• SME25	**6SL3055-0AA00-5HA3**	–	O	O	O	O	O	O
• SME120	**6SL3055-0AA00-5JA3**	–	O	O	O	O	O	O
• SME125	**6SL3055-0AA00-5KA3**	–	O	O	O	O	O	O
SINAMICS S120, supplementary system components:	See SiePortal							
• DMC20 DRIVE-CLiQ Hub Module	**6SL3055-0AA00-6AA1**	–	O	O	O	O	O	O
• DME20 DRIVE-CLiQ Hub Module	**6SL3055-0AA00-6AB0**	–	O	O	O	O	O	O
SINAMICS S120 Combi	See SINAMICS S120 drive system	–	O	O	O	O	O	O
SINAMICS S120 Motor Modules Booksize format:	See SINAMICS S120 drive system							
• Internal air cooling	6SL3120-...	–	O	O	O	O	O	O
SINAMICS S120 Active Line Modules Booksize format:	See SINAMICS S120 drive system							
• Internal air cooling	6SL3130-...	–	O	O	O	O	O	O
SINAMICS S120 Smart Line Modules Booksize format:	See SINAMICS S120 drive system							
• Internal air cooling	6SL3130-...	–	O	O	O	O	O	O
SINAMICS S120 Motor Modules Chassis format Internal air cooling	Rated pulse frequency 2 kHz On request	–	O	O	O	O	O	O
SINAMICS S220 Motor Modules Booksize format:	See SINAMICS S220 drive system							
• Internal air cooling	6SL5120-...	–	O	O	O	O	O	O
SINAMICS S220 Smart Line Modules Booksize format:	See SINAMICS S220 drive system							
• Internal air cooling	6SL5130-...	–	O	O	O	O	O	O

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Connectable measuring systems

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Connectable measuring systems								
Number of measuring systems per axis, maximum	Not valid for PLC drives	2	2	2	2	2	2	2
Absolute encoders installed in the motors:								
• SIMOTICS S-1FK2		–	O	O	O	O	O	O
• SIMOTICS S-1FL2		O	–	–	–	–	–	–
• SIMOTICS S-1FT2		–	O	O	O	O	O	O
• SIMOTICS S-1FK7		–	O	O	O	O	O	O
• SIMOTICS S-1FT7		–	O	O	O	O	O	O
• SIMOTICS M-1PH3 Pro		O	–	–	–	–	–	–
• SIMOTICS M-1PH8		–	O	O	O	O	O	O
Incremental encoders installed in the motors:								
• SIMOTICS S-1FK7		–	O	O	O	O	O	O
• SIMOTICS S-1FT7		–	O	O	O	O	O	O
• SIMOTICS M-1PH3 Pro		O	O	O	O	O	O	O
• SIMOTICS M-1PH8		–	O	O	O	O	O	O
Resolvers installed in SIMOTICS S-1FK7 motors		–	O	O	O	O	O	O
Rotary measuring systems with:	See Motion Control Encoder measuring systems via SINAMICS Sensor Modules SMC							
• sin/cos 1 V_{pp}	6FX2001-3...	–	O	O	O	O	O	O
• RS422 (TTL)	6FX2001-2...	–	O	O	O	O	O	O
• Distance-coded reference marks		–	O	O	O	O	O	O
• EnDat 2.1/EnDat 2.2		–	O	O	O	O	O	O
• DRIVE-CLiQ	6FX2001-5...	–	O	O	O	O	O	O
Linear measuring systems LMS with:	See SiePortal							
• sin/cos 1 V_{pp}		–	O	O	O	O	O	O
• Distance-coded reference marks RS422 (TTL)		–	O	O	O	O	O	O
• EnDat 2.1		–	O	O	O	O	O	O
Absolute encoder connection with SSI interface		–	O	O	O	O	O	O
Certified DQI encoders (DQI 100)	See Motion Control Encoder measuring systems	–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Connectable motors								
SIMOTICS S-1FK2 servomotor	(preferred version)	–	O	O	O	O	O	O
SIMOTICS S-1FT2 servomotor		–	O	O	O	O	O	O
SIMOTICS S-1FK7 servomotor		–	O	O	O	O	O	O
SIMOTICS S-1FT7 servomotor		–	O	O	O	O	O	O
SIMOTICS S-1FL2 servomotor		O	–	–	–	–	–	–
SIMOTICS M-1PH3 Pro spindle motor for SINAMICS S200 Spindle		O	–	–	–	–	–	–
SIMOTICS M-1PH3 Pro spindle motor	(preferred version)	–	O	O	O	O	O	O
SIMOTICS M-1PH8 spindle motor		–	O	O	O	O	O	O
SIMOTICS M-1FE1/1FE2 built-in motor		–	O	O	O	O	O	O
SIMOTICS L-1FN3 linear motor		–	O	O	O	O	O	O
SIMOTICS T-1FW6 built-in torque motor		–	O	O	O	O	O	O
Hybrid spindle/motor spindle 2SP1	See SiePortal	–	O	O	O	O	O	O
Non-Siemens motor	On request	–	O	O	O	O	O	O

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Overview of functions
SINUMERIK 828D and SINUMERIK 828D Smart CNCs

Axis functions

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
Axis functions								
Feedrate override		0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %
Feedrate override, axis-specific		0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %
Traversing range, decades		± 9	± 9	± 9	± 9	± 9	± 9	± 9
Rotary axis, turning endlessly		✓	✓	✓	✓	✓	✓	✓
Velocity, maximum		300 m/s	300 m/s	300 m/s	300 m/s	300 m/s	300 m/s	300 m/s
Acceleration with jerk limitation		✓	✓	✓	✓	✓	✓	✓
Programmable acceleration		✓	✓	✓	✓	✓	✓	✓
Follow-up mode		✓	✓	✓	✓	✓	✓	✓
Measuring systems 1 and 2, selectable		✓	✓	✓	✓	✓	✓	✓
Feedrate interpolation		✓	✓	✓	✓	✓	✓	✓
Separate path feedrate for roundings and chamfers		✓	✓	✓	✓	✓	✓	✓
Travel to fixed stop		✓	✓	✓	✓	✓	✓	✓
Jerk adaptation	**6FC5800-0AS22-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Travel to fixed stop with Force Control	**6FC5800-0AM01-0YH0**	–	O	O	O	O	O	O
Analog axes/spindles								
• Basic configuration		1	1	1	1	1	1	1
• Maximum configuration via analog axis module (retrofit solution) with 2 analog axes per HLA module and 1 axis/spindle on the PPU/PU	For more information, go to: https://support.industry.siemens.com/cs/ww/en/view/109475900	1	5	5	5	5	5	5
Analog axis module (retrofit solution)	**6FC5300-0BA01-0AA0** Limited function – only for analog axis	–	2	2	2	2	2	2
Setpoint exchange:	**6FC5800-0AM05-0YH0**							
• Turning		–	–	–	–	–	O	O
• Milling		–	–	–	–	–	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	O	O
Tangential control:	**6FC5800-0AM06-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Position switching signals/cam controller		–	–	–	–	–	–	–
Advanced Position Control APC ECO	**6FC5800-0AM12-0YH0**	–	O	O	O	O	O	O
Advanced Position Control APC		–	–	–	–	–	–	–
Simulated motor encoder	**6FC5800-0AD30-0YH0**	–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Spindle functions								
Spindle speed, analog setpoint	Connection to PPU/PU See analog axes/spindles	1	1	1	1	1	1	1
Spindle speed, digital setpoint		✓	✓	✓	✓	✓	✓	✓
Spindle speed, max. programmable value range Display: ± 999999999.9999		10^6 ... 10^{-4}	10^6 ... 10^{-4}	10^6 ... 10^{-4}	10^6 ... 10^{-4}	10^6 ... 10^{-4}	10^6 ... 10^{-4}	10^6 ... 10^{-4}
Spindle override		0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %	0 ... 200 %
Gear stages		5	5	5	5	5	5	5
Intermediate gear		✓	✓	✓	✓	✓	✓	✓
Gear stage selection, automatic		✓	✓	✓	✓	✓	✓	✓
Oriented spindle stop		✓	✓	✓	✓	✓	✓	✓
Spindle speed limitation min./max.		✓	✓	✓	✓	✓	✓	✓
Constant cutting speed		✓	✓	✓	✓	✓	✓	✓
Spindle control via PLC (positioning, oscillation)		✓	✓	✓	✓	✓	✓	✓
Changeover to axis mode		✓	✓	✓	✓	✓	✓	✓
Axis synchronization on-the-fly		✓	✓	✓	✓	✓	✓	✓
Thread run-in and run-out, programmable		✓	✓	✓	✓	✓	✓	✓
Thread cutting with constant or variable pitch		✓	✓	✓	✓	✓	✓	✓
Tapping with compensating chuck/rigid tapping		✓	✓	✓	✓	✓	✓	✓

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Interpolations

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
Interpolations								
Linear interpolating axes, maximum simultaneously interpolating		4	4	4	4	4	4	4
Circle via center point and end point		✓	✓	✓	✓	✓	✓	✓
Circle via intermediate point		✓	✓	✓	✓	✓	✓	✓
Helical interpolation		✓	✓	✓	✓	✓	✓	✓
Universal interpolator NURBS (non-uniform rational B splines)		✓	✓	✓	✓	✓	✓	✓
Continuous-path mode with programmable rounding clearance		✓	✓	✓	✓	✓	✓	✓
Multi-axis interpolation > 4 interpolating axes		–	–	–	–	–	–	–
Advanced Surface:	6FC5800-0AS07-0YH0							
• Turning		–	–	–	–	–	–	–
• Milling		✓	✓	✓	✓	✓	✓	✓
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Top Surface:	**6FC5800-0AS17-0YH0** Requirement: Advanced Surface							
• Turning		–	–	–	–	–	–	–
• Milling		O	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Optimized Top Surface [1]	NEW **6FC5800-0AH00-0YH0** Requirement: Top Surface							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Optimized IPO cycle	NEW **6FC5800-0AH01-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	O	O
Top Speed plus	**6FC5800-0AS62-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Surface turning	**6FC5800-0AR51-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	O	O	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Nodding compensation ECO • One compensation axis can process 1 influence quantity • Adaptive compensation via 3 interpolation points	**6FC5800-0AS20-0YH0** No rotary axis possible	–	O	O	O	O	O	O
Nodding compensation ADVANCED • Each compensation axis can process 3 influence quantities • Adaptive compensation via 3 interpolation points • Unlimited number of compensation axes	**6FC5800-0AS21-0YH0** No rotary axis possible	–	O	O	O	O	O	O

[1] Not released in combination with Protect MyMachine /3D Primitives (option S03).

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Interpolations (continued)								
Path acceleration limitation	**6FC5800-0AP26-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Spline interpolation (A, B and C splines)	**6FC5800-0AS16-0YH0**	–	O	O	O	O	O	O
Compressor for 3-axis machining COMPCAD:								
• Turning	Without CYCLE832	✓	✓	✓	✓	✓	✓	✓
• Milling		✓	✓	✓	✓	✓	✓	✓
• G-Tech Cylindrical/G-Tech Surface	Without CYCLE832	✓	✓	✓	✓	✓	✓	✓
Polynomial interpolation		–	–	–	–	–	–	–
Involute interpolation		–	–	–	–	–	–	–
Crank interpolation CRIP		–	–	–	–	–	–	–
Interpolation turning	**6FC5800-0AP57-0YH0**							
• Turning		–	–	–	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE cone turning [1] Requirement: CNC software ≥ SW version 4.95	**6FC5800-0AR52-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE QR Code [1] Requirement: CNC software ≥ SW version 4.95 SP3	**6FC5800-0AR53-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE Engraving [1] Requirement: CNC software ≥ SW version 4.95 SP3	**6FC5800-0AR54-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Prime Turning ™	**6FC5800-0AR56-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE Chatter avoidance [1] NEW Requirement: CNC software ≥ SW version 4.95 SP4	**6FC5800-0AR57-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–

[1] Regarding delivery, please contact your local Siemens sales office or Regional Company.
Requirement:
Execution from external storage EES, Article No. 6FC5800-0AP75-0YH0, or CNC user memory expanded for programs, Article No. 6FC5800-0AP77-0YH0.

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Interpolations

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Interpolations (continued)								
SMTE Trochoidal Basic [1] NEW Requirement: CNC software ≥ SW version 4.95 SP4	6FC5800-0AR58-0YH0							
• Turning		–	–	–	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE Eccentric Basic [1] NEW Requirement: CNC software ≥ SW version 4.95 SP4	6FC5800-0AR60-0YH0							
• Turning		–	–	–	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	O	O	O	O
SMTE Eccentric Advanced [1] NEW Requirement: CNC software ≥ SW version 4.95 SP4	6FC5800-0AR61-0YH0							
• Turning		–	–	–	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	O	O	O	O
Continuous path mode (G64) extended NEW	6FC5800-0AS37-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Comb grooving	6FC5800-0AS58-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Turning with Y-axis	6FC5800-0AS56-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Advanced Rapid Movement (COMPPATH) Intelligent adaptation of the machine paths outside of the workpiece	6FC5800-0AS64-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Reduced dynamic mode	6FC5800-0AS65-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O

[1] Regarding delivery, please contact your local Siemens sales office or Regional Company.
Requirement:
Execution from external storage EES, Article No. 6FC5800-0AP75-0YH0, or CNC user memory expanded for programs, Article No. 6FC5800-0AP77-0YH0.

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Couplings								
Pair of synchronized axes (gantry axes), Basic:	**6FC5800-0AS51-0YH0**	O	O	O	O	O	O	O
• Turning		1	1	1	1	1	1	1
• Milling		1	1	1	1	1	2	2
• G-Tech Cylindrical/G-Tech Surface		1	1	1	1	1	1	1
Main axis/sub axis Basic for drives	**6FC5800-0AS52-0YH0**	–	O	O	O	O	O	O
• Turning		–	1	1	1	1	2	2
• Milling		–	1	1	1	1	3	3
• G-Tech Cylindrical/G-Tech Surface		–	1	1	1	1	2	2
Generic couplings:								
CP-Static, e.g. counterspindle:	**6FC5800-0AM75-0YH0**							
• 1 × basic synchronous spindle, coupling ratio 1:1, no multi-edge machining:								
- Turning		–	–	–	–	–	O	O
- Milling		–	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
CP-Basic, e.g. multi-edge turning:	**6FC5800-0AM72-0YH0**							
• 4 axis pairs in simultaneous coupled motion:								
- Turning		O	O	O	O	O	O	O
- Milling		–	–	–	–	–	O	O
- G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
• 1 × synchronous spindle/multi-edge turning:								
- Turning		O	O	O	O	O	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
• Master value coupling/curve table interpolation		–	–	–	–	–	–	–
CP-Comfort, e.g. electronic gear:	**6FC5800-0AM73-0YH0**							
• 4 axis pairs in simultaneous coupled motion:								
- Turning		–	O	O	O	O	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
• 1 × synchronous spindle/multi-edge turning:								
- Turning		–	O	O	O	O	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
• Electronic gear for 3 leading axes, without curve table, without cascading:								
- Turning		–	O	O	O	O	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
• Axial coupling in the machine coordinate system		–	–	–	–	–	–	–
• Master value coupling/curve table interpolation		–	–	–	–	–	–	–

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Transformations

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Transformations								
Cartesian point-to-point travel PTP		✓	✓	✓	✓	✓	✓	✓
TRANSMIT/cylinder surface transformation	**6FC5800-0AM27-0YH0**	–	O	O	O	O	O	O
TRANSMIT/TRACYL transformation without Y axis	**6FC5800-0AS50-0YH0**	O	O	O	O	O	O	O
Inclined axis:	**6FC5800-0AM28-0YH0**							
• Turning	For non-orthogonal Y axis	–	–	–	O	O	O	O
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical	For non-orthogonal X axis	–	O	O	O	O	O	O
• G-Tech Surface		–	–	–	–	–	–	–
Inclined axis Basic, fixed angle:	**6FC5800-0AS54-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface	For non-orthogonal X axis	–	O	O	O	O	O	O
Simple oscillation function, modal:								
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
Concatenated transformations inclined axis TRAANG after cardanic milling head/TRANSMIT/TRACYL:								
• Turning		–	–	–	✓	✓	✓	✓
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical		–	✓	✓	✓	✓	✓	✓
• G-Tech Surface		–	–	–	–	–	–	–

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Measuring functions and measuring cycles								
Measuring stage 1 Two probes switching with/without deletion of distance-to-go		✓	✓	✓	✓	✓	✓	✓
Measuring stage 2 Axial measurement, measurements from synchronized actions, cyclic measurement, fast measurement	**6FC5800-0AM32-0YH0**							
• Turning		–	–	–	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	O	O	O	O
Measuring cycles for drilling/milling and turning Calibration of workpiece probe, workpiece measurement, tool measurement:	**6FC5800-0AP28-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Measure kinematics Determine transformation data of rotary axes:	**6FC5800-0AP18-0YH0**							
• Turning		–	–	–	–	–	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Logging of measurement results		–	✓	✓	✓	✓	✓	✓
Technologies								
Handwheel override		✓	✓	✓	✓	✓	✓	✓
Kinematic chain		✓	✓	✓	✓	✓	✓	✓
Contour handwheel	**6FC5800-0AM08-0YH0**	O	O	O	O	O	O	O
Angle head adapter	**6FC5800-0AM56-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Multiple feedrates in one block:								
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
Continuous dressing, parallel dressing Online modification of the tool offset:								
• Turning/milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
Balance cutting:	**6FC5800-0AS05-0YH0**							
• Turning		–	–	–	–	–	O	O
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
Technologies (continued)								
Oscillation function, non-modal, modal and asynchronous:	**6FC5800-0AM34-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface	CYCLE4071 ... CYCLE4079	O	O	O	O	O	O	O
Extended functions for grinding Grinding advanced:	**6FC5800-0AS35-0YH0**							
• Dressing (paraxial form-truing) with stock removal cycles (Basic CYCLE95)								
• Cylinder error compensation								
- Turning		–	–	–	–	–	–	–
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
SINUMERIK MDynamics:								
• Advanced Surface:	6FC5800-0AS07-0YH0							
- Turning		–	–	–	–	–	–	–
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Top Surface:	**6FC5800-0AS17-0YH0**							
- Turning		–	–	–	–	–	–	–
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• High Speed Settings:								
- Turning		–	–	–	–	–	–	–
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SINUMERIK 828D tool orientation (TRAORI, RTCP)	**6FC5800-0AM30-0YH0**							
• Turning		–	–	–	–	–	O	O
• Milling		–	–	–	–	–	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Top Speed plus	**6FC5800-0AS62-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Easy connection of robots and handling systems Run MyRobot /EasyConnect		✓	✓	✓	✓	✓	✓	✓
Run MyRobot /Handling	**6FC5800-0AP72-0YH0**	–	O	O	O	O	O	O
Optimize MyMachining /AdaptiveControl Standard	**6FC5800-0AT00-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Optimize MyMachining /AdaptiveControl Pro	**6FC5800-0AT02-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Technologies (continued)								
ACM Suite: OMM /ToolMonitor Pro	**6FC5800-0AT03-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
ACM Suite: Multi-spindle extension	**6FC5800-0AT04-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
ACM Suite: Multi-channel extension	**6FC5800-0AT05-0YH0** **MCS2154**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
ACM Suite: Single part extension	**6FC5800-0AT06-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O

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Motion-synchronous actions

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
Motion-synchronous actions								
CNC inputs/outputs, high-speed:								
• Digital inputs drives onboard		–	12	12	12	12	12	12
• Digital inputs or outputs drives onboard, parameterizable		–	8	8	8	8	8	8
• Digital inputs CNC onboard		48	8	8	8	8	8	8
• Digital outputs CNC onboard		32	8	8	8	8	8	8
Synchronized actions and fast auxiliary function output incl. 3 synchronous functions		✓	✓	✓	✓	✓	✓	✓
Positioning axes and spindles via synchronized actions (command axes)		✓	✓	✓	✓	✓	✓	✓
Analog value control in the IPO cycle		–	–	–	–	–	–	–
Evaluation of internal drive variables, Basic	**6FC5800-0AS53-0YH0**	–	O	O	O	O	O	O
Asynchronous subprograms ASUB		✓	✓	✓	✓	✓	✓	✓
Interrupt routines with fast retraction from the contour (with subprogram/ASUB/LIFTFAST)		✓	✓	✓	✓	✓	✓	✓
Cross-mode actions (ASUBs and synchronized actions in all operating modes)		✓	✓	✓	✓	✓	✓	✓
Online tool correction:								
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
Display active synchronized actions in HMI:	Included in option: Extended operator functions 6FC5800-0AP16-0YH0							
• Turning		O	O	O	O	O	O	O
• Milling		O	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
Engineering								
Program screens, operating areas and user interfaces **Create MyHMI /3GL:**	OEM contract required							
Program screens, operating areas, and user interfaces in high-level languages 3 languages (en, de, zh)								
• Single license and for download, without data storage medium	**MCS1200**	–	O[1]	O	O[1]	O	O[1]	O
Use extended HMI applications Run MyHMI /3GL (maximum 300 MB)	**6FC5800-0AP60-0YH0**	–	O	O	O	O	O	O
Integrate screens in SINUMERIK Operate **Run MyScreens**								
• Free screens		5	5	5	5	5	5	5
• > 5 screens, extended functions:	**6FC5800-0AP64-0YH0**							
- Turning		O	O	O	O	O	O	O
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
SINUMERIK 828D Run EasyXML:								
• Turning		✓	✓	✓	✓	✓	✓	✓
• Milling		✓	✓	✓	✓	✓	✓	✓
• G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
Analyze MyMachine /SINUMERIK Tuning Center Analyze MyMachine /SINUMERIK Tuning Center (incl. Expert Trace) Software (without license) Software for path and axis optimization of a machine tool with SINUMERIK 828 including Diagnostic Trace Requirement: SINUMERIK Operate on PC, Operate Commissioning tool (supplied on Toolbox DVD-ROM or download version, Article No. 6FC5830-0CY50-0YA8 or 6FC5830-0EY50-0YA8)	**6FC5862-4YC67-0YA0**	–	O	O	O	O	O	O
Development, commissioning and simulation of virtual machine tools with SINUMERIK 828								
Create MyVirtual Machine /Operate Virtual test environment for SINUMERIK 828	NEW **MCS4250f** [3] **MCS4250n** [4] **MCS4250a** [5]	O[2]	O[2]	O[2]	O[2]	O[2]	O[2]	O[2]
Create MyVirtual Machine /Open Virtual test environment for SINUMERIK 828 Extension with an open communications interface	NEW **MCS4255f** [3] **MCS4255n** [4] **MCS4255a** [5]	O[2]	O[2]	O[2]	O[2]	O[2]	O[2]	O[2]
Create MyVirtual Machine /3D Virtual test environment for SINUMERIK 828 Extension comprising a Virtual Machine Creator for JobShop applications	NEW **MCS4210** [3] **MCS4210n** [4] **MCS4210a** [5]	O[2]	O[2]	O[2]	O[2]	O[2]	O[2]	O[2]

[1] Applies exclusively to programming the Embedded HMI with Qt. You can obtain the Qt license key directly from: The Qt Company: https://www.qt.io

[2] In Create MyVirtual Machine, no distinctions are mapped between the hardware types/classes. Projects created with Create/Run MyVirtual Machine can be loaded into SINUMERIK 828 PU/PPU, taking into consideration the HW configurations and constraints.

[3] Floating: Server-based licenses, can be used on multiple work stations (not simultaneously); order recommendation.

[4] Node-locked Server: Server license that is connected to particular clients (work station/PC)
Node-locked Standalone: Local license file that is connected to a particular client (work station/PC)

[5] Academics: For non-commercial use.

For more information, request the detailed description of licenses.

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CNC programming language

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
CNC programming language								
Programming language DIN 66025 and high-level language expansion		✓	✓	✓	✓	✓	✓	✓
Main program call from main program and subprogram		✓	✓	✓	✓	✓	✓	✓
Subprogram levels, maximum		11	11	11	11	11	11	11
Interrupt routines, maximum		4	4	4	4	4	4	4
Number of subprogram passes		≤ 9999	≤ 9999	≤ 9999	≤ 9999	≤ 9999	≤ 9999	≤ 9999
Number of levels for skip blocks		2	2	2	2	2	2	2
Number of levels for skip blocks, maximum:	Included in option: Extended operator functions 6FC5800-0AP16-0YH0	10	10	10	10	10	10	10
• Turning		O	O	O	O	O	O	O
• Milling		O	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
Polar coordinates		✓	✓	✓	✓	✓	✓	✓
1/2/3-point contours		✓	✓	✓	✓	✓	✓	✓
Dimensions metric/inch, changeover via operator action or program		✓	✓	✓	✓	✓	✓	✓
Inverse-time feedrate		✓	✓	✓	✓	✓	✓	✓
Auxiliary function output via:								
• M word, max. programmable value range: INT 2^{31} -1 ... 2^{31}		✓	✓	✓	✓	✓	✓	✓
• H word, max. programmable value range: REAL ± 3.4028 ex38, INT -2^{31} ... 2^{31} -1	Display: ± 999999999,9999	✓	✓	✓	✓	✓	✓	✓

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
CNC programming language (continued)								
CNC high-level language with:								
• User variables, GUD, configurable		✓	✓	✓	✓	✓	✓	✓
• Predefined user variables (R-parameters), commentable		3000	3000	3000	3000	3000	3000	3000
• Predefined global user variables (global R parameters), commentable:								
- Turning		–	–	–	–	–	100	100
- Milling		–	–	–	–	–	100	100
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	100	100
• Predefined user variables LUD (R parameters), configurable		✓	✓	✓	✓	✓	✓	✓
• Read/write system variables		✓	✓	✓	✓	✓	✓	✓
• Indirect programming		✓	✓	✓	✓	✓	✓	✓
• Program jumps and branches		✓	✓	✓	✓	✓	✓	✓
• Dynamic jumps RETB		✓	✓	✓	✓	✓	✓	✓
• Program coordination with WAIT, START, INIT:								
- Turning		–	–	–	–	–	✓	✓
- Milling		–	–	–	–	–	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	✓	✓
• Arithmetic and trigonometric functions		✓	✓	✓	✓	✓	✓	✓
• Comparison operations and logic combinations		✓	✓	✓	✓	✓	✓	✓
• Macro techniques		✓	✓	✓	✓	✓	✓	✓
• Control structures IF-ELSE-ENDIF		✓	✓	✓	✓	✓	✓	✓
• Control structures WHILE, FOR, REPEAT, LOOP		✓	✓	✓	✓	✓	✓	✓
• Commands to HMI		✓	✓	✓	✓	✓	✓	✓
• STRING functions		✓	✓	✓	✓	✓	✓	✓

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CNC programming language

Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
CNC programming language (continued)								
Program functions:								
• Dynamic preprocessing buffer (FIFO)		✓	✓	✓	✓	✓	✓	✓
• Look Ahead, recorded part program blocks:								
- Turning	Without CYCLE832	150	150	150	300	300	450	450
- Milling with MDynamics Advanced Surface	COMPCAD with CYCLE832	150	150	150	300	300	450	450
- Milling with MDynamics Top Surface	COMPSURF with CYCLE832	2000	2000	2000	2000	2000	2000	2000
- G-Tech Cylindrical	Without CYCLE832	150	150	150	300	300	450	450
- G-Tech Surface	Without CYCLE832	150	150	150	300	300	450	450
• Look Ahead, IPO blocks, buffered:								
- Turning	Max. 150 without CYCLE832	50	50	50	100	100	150	150
- Milling with MDynamics Advanced Surface	COMPCAD with CYCLE832	50	50	50	100	100	150	150
- Milling with MDynamics Top Surface	COMPSURF with CYCLE832	200	200	200	200	200	200	200
- G-Tech Cylindrical	Max. 150 without CYCLE832	50	50	50	100	100	150	150
- G-Tech Surface	Max. 150 without CYCLE832	50	50	50	100	100	150	150
• Frame concept		✓	✓	✓	✓	✓	✓	✓
• Inclined-surface machining with swivel cycle:								
- Turning		–	–	–	–	–	–	–
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
• Axis/spindle replacement		✓	✓	✓	✓	✓	✓	✓
• Geometry axes, switchable online in the CNC program		✓	✓	✓	✓	✓	✓	✓
• Program preprocessing		✓	✓	✓	✓	✓	✓	✓
Online ISO dialect interpreter:								
• Turning		✓	✓	✓	✓	✓	✓	✓
• Milling		✓	✓	✓	✓	✓	✓	✓
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
CNC programming language (continued)								
Program/workpiece management:								
• Part programs on PPU/PU, maximum number Only limited by the available memory with the following options: Execution from external storage EES (6FC5800-0AP75-0YH0) or expansion of the CNC user memory (6FC5800-0AP77-0YH0)	In total a maximum of 512 files per directory	750	750	750	750	750	750	750
• Workpieces on PPU/PU, maximum number Only limited by the available memory with the following options: Execution from external storage EES (6FC5800-0AP75-0YH0) or expansion of the CNC user memory (6FC5800-0AP77-0YH0)	In total a maximum of 256 directories	250	250	250	250	250	250	250
• On USB storage medium, e.g. USB flash drive	Hard disk not possible	✓	✓	✓	✓	✓	✓	✓
• On network drive (Windows Share/FTP)		✓	✓	✓	✓	✓	✓	✓
• Templates for workpieces, programs and INI files		✓	✓	✓	✓	✓	✓	✓
• Job lists		✓	✓	✓	✓	✓	✓	✓
Basic frames, maximum number		1	1	1	1	1	1	1
Settable offsets, maximum number		50	50	50	100	100	100	100
Work offsets, programmable (frames)		✓	✓	✓	✓	✓	✓	✓
Work offsets, fit-dependent:	Fit-dependent corrections							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
Scratching, determining work offset		✓	✓	✓	✓	✓	✓	✓
Work offsets, external via PLC		✓	✓	✓	✓	✓	✓	✓
Global and local user data		✓	✓	✓	✓	✓	✓	✓
Global program user data		✓	✓	✓	✓	✓	✓	✓
Display and log system variables also via online configurable display		–	–	–	–	–	–	–
Programming support								
Program editor:								
• Programming support for cycles, programGUIDE		✓	✓	✓	✓	✓	✓	✓
• CNC editor with editing functions: select, copy, delete		✓	✓	✓	✓	✓	✓	✓
• Geometry processor with programming graphics/free contour input (contour calculator)		✓	✓	✓	✓	✓	✓	✓
• Screens for 1/2/3-point contours (contour definition programming)		–	–	–	–	–	–	–
• **ShopTurn/ShopMill machining step programming:**	**6FC5800-0AP17-0YH0**							
- Turning		O	O	O	O	O	O	O
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• **programSYNC – multi-channel step sequence programming:**	**6FC5800-0AP05-0YH0**							
- Turning		–	–	–	–	–	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–

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Programming support

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Programming support (continued)								
• Manual machine functions:	Included in option: ShopTurn/ShopMill machining step programming 6FC5800-0AP17-0YH0							
- Turning		O	O	O	O	O	O	O
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Backup of workpiece setup data:	Included in option: Extended operator functions 6FC5800-0AP16-0YH0							
- Turning		O	O	O	O	O	O	O
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
• Multiple clamping of various workpieces:	Included in option: ShopTurn/ShopMill machining step programming 6FC5800-0AP17-0YH0							
- Turning		–	–	–	–	–	–	–
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Technology cycles for drilling/milling:	Basic scope							
• Turning		✓	✓	✓	✓	✓	✓	✓
• Milling		✓	✓	✓	✓	✓	✓	✓
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Technology cycles for turning:	Basic scope							
• Turning		✓	✓	✓	✓	✓	✓	✓
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Technology cycles for grinding Grinding basic		–	–	–	–	–	–	–

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Programming support (continued)								
Advanced technology functions (expansion of the technology cycles for turning and milling):	**6FC5800-0AP58-0YH0**							
• Pocket milling with free contour definition and islands:								
- Turning		O	O	O	✓	✓	✓	✓
- Milling		O	O	O	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Stock removal cycles with free contour definition:								
- Turning		O	O	O	✓	✓	✓	✓
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Residual material detection and machining for contour pockets and stock removal:	**6FC5800-0AP13-0YH0** Requirement: Advanced technology functions option 6FC5800-0AP58-0YH0							
• Turning		O	O	O	O	O	O	O
• Milling		O	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Access protection for cycles Lock MyCycles	**6FC5800-0AP54-0YH0**	O	O	O	O	O	O	O
Programming support can be extended, e.g. customer cycles		✓	✓	✓	✓	✓	✓	✓
DXF Reader for PC integrated in SINUMERIK Operate	**6FC5800-0AP56-0YH0**	–	O	O	O	O	O	O
Monitoring for maximum tool speed/acceleration	**6FC5800-0AS08-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Configured stop	**6FC5800-0AS24-0YH0**	–	O	O	O	O	O	O

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Simulations

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
Simulations								
Simulation of program X, while program Y is being executed (simulation parallel to machining)		–	–	–	–	–	–	–
Simulation (finished part) in 2D representation		✓	✓	✓	✓	✓	✓	✓
3D simulation 1 (finished part):	**6FC5800-0AP25-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Simulation of finished part and working area in 3D representation		–	–	–	–	–	–	–
Simulation of finished part with collision check in 3D representation		–	–	–	–	–	–	–
Simultaneous recording (real-time simulation of the current machining operation)	**6FC5800-0AP22-0YH0**	O	O	O	O	O	O	O
Simulation of virtual machine tools with SINUMERIK 828 during production								
Run MyVirtual Machine /Operate NEW Control-identical programming station for NC programs or for training and apprenticeship purposes	**MCS4220** [1] **MCS4220n** [2] **MCS4220a** [3]	O	O	O	O	O	O	O
Run MyVirtual Machine /Open NEW Extension with an open communications interface for integration into a simulation system	**MCS4230** [1] **MCS4230n** [2] **MCS4230a** [3]	O	O	O	O	O	O	O
Run MyVirtual Machine /3D NEW Extension to the virtual machine during work preparation	**MCS4240** [1] **MCS4240n** [2] **MCS4240a** [3]	O	O	O	O	O	O	O

[1] Floating: Server-based licenses, can be used on multiple workstations (not simultaneously); order recommendation.

[2] Node-locked Server: Server license that is connected to particular clients (work station/PC)
Node-locked Standalone: Local license file that is connected to a particular client (work station/PC)

[3] Academics: For non-commercial use.

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Operating modes								
JOG:								
• Handwheel selection		✓	✓	✓	✓	✓	✓	✓
• Inch/metric changeover		✓	✓	✓	✓	✓	✓	✓
• Manual measurement of work offset:								
- Turning		✓	✓	✓	✓	✓	✓	✓
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Additional measuring version beyond the standard scope: Standard scope workpiece zero: Set edge, align edge, right-angled corner, 1 hole, 1 circular spigot and rectangular spigot Expansion of the measurement screens via combo box	Included in option: Extended operator functions 6FC5800-0AP16-0YH0							
- Turning		–	–	–	–	–	–	–
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Additional measuring version beyond the standard scope: Standard scope workpiece zero: Set edge, 1 circular spigot and rectangular spigot Expansion of the measurement screens via combo box	NEW Included in option: Extended operator functions 6FC5800-0AQ06-0YH0	O	✓	✓	✓	✓	✓	✓
• Manual measurement of tool offset:								
- Turning		✓	✓	✓	✓	✓	✓	✓
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface	Manual dresser setup	✓	✓	✓	✓	✓	✓	✓
• Automatic tool/workpiece measurement:								
- Turning		✓	✓	✓	✓	✓	✓	✓
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface	Manual workpiece setup	✓	✓	✓	✓	✓	✓	✓
• Reference point approach, automatic/via CNC program		✓	✓	✓	✓	✓	✓	✓
MDI:								
• Input in text editor		✓	✓	✓	✓	✓	✓	✓
• Load/save MDI program:	Included in option: Extended operator functions 6FC5800-0AP16-0YH0							
- Turning		O	O	O	O	O	O	O
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
• Input screen forms for technology and positioning, cycle support		✓	✓	✓	✓	✓	✓	✓

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Operating modes

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
Operating modes (continued)								
Teach-in:	Included in option: Extended operator functions 6FC5800-0AP16-0YH0							
• Turning		O	O	O	O	O	O	O
• Milling		O	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
Automatic:								
• Execution from storage medium on the front USB interface of the operator panel, e.g. card reader, USB flash drive	Hard disk not possible	✓	✓	✓	✓	✓	✓	✓
• Execution from storage medium on the rear USB interface of the operator panel, e.g. card reader, USB flash drive	Hard disk not possible	✓	✓	✓	✓	✓	✓	✓
• Execution from network drive		✓	✓	✓	✓	✓	✓	✓
• Program control		✓	✓	✓	✓	✓	✓	✓
• Program editing		✓	✓	✓	✓	✓	✓	✓
• Overstore:	Included in option: Extended operator functions 6FC5800-0AP16-0YH0							
- Turning		O	O	O	O	O	O	O
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
• DRF offset:	Included in option: Extended operator functions 6FC5800-0AP16-0YH0							
- Turning		O	O	O	O	O	O	O
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
• Block search with/without calculation		✓	✓	✓	✓	✓	✓	✓
• Extended block search Program, search pointer, step up and down, interrupt function:	Included in option: Extended operator functions 6FC5800-0AP16-0YH0							
- Turning		O	O	O	O	O	O	O
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
Repos (repositioning on the contour):	NEW Included in option: Extended operator functions 6FC5800-0AQ06-0YH0							
• Via operator input/semi-automatically		O	✓	✓	✓	✓	✓	✓
• Program		O	✓	✓	✓	✓	✓	✓
Preset:								
• Set actual value		✓	✓	✓	✓	✓	✓	✓

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Tools								
Tool types:								
• Turning:								
- Turning		✓	✓	✓	✓	✓	✓	✓
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Drilling/milling:								
- Turning		✓	✓	✓	✓	✓	✓	✓
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Groove sawing:								
- Turning		✓	✓	✓	✓	✓	✓	✓
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Grinding/dressing:								
- Turning		–	–	–	–	–	–	–
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
• Multi-tool:								
- Turning		–	✓	✓	✓	✓	✓	✓
- Milling		–	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Tool radius compensations in plane with:								
• Approach and retract strategies		✓	✓	✓	✓	✓	✓	✓
• Transition circle/ellipse on outer edges		✓	✓	✓	✓	✓	✓	✓
Configurable intermediate blocks with tool radius compensation active		✓	✓	✓	✓	✓	✓	✓
Tool radius compensation in 3D representation		–	–	–	–	–	–	–
Tool carrier with orientation capability:								
• Turning		–	–	–	✓	✓	✓	✓
• Milling		✓	✓	✓	✓	✓	✓	✓
• G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
Look-ahead detection of contour violations		✓	✓	✓	✓	✓	✓	✓
Constant grinding wheel peripheral speed GWPS:								
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓

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Tool management

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
Tool management								
Operation with tool management:								
• Real magazines, maximum number:	NEW Included in option: Extended operator functions 6FC5800-0AQ06-0YH0							
- Turning		O	1	1	1	1	2	2
- Milling		O	1	1	1	1	2	2
- G-Tech Cylindrical/G-Tech Surface		O	1	1	2	2	2	2
• Tool list		✓	✓	✓	✓	✓	✓	✓
• Expandable tool list		–	–	–	–	–	–	–
• Tools in tool list:								
- Turning		128	128	128	256	256	768	768
- Milling		128	128	128	256	256	768	768
- G-Tech Cylindrical/G-Tech Surface		128	128	128	256	256	768	768
• Cutting edges in tool list:								
- Turning		256	256	256	512	512	1536	1536
- Milling		256	256	256	512	512	1536	1536
- G-Tech Cylindrical/G-Tech Surface		256	256	256	512	512	1536	1536
• Tool offset selection via T and D numbers		✓	✓	✓	✓	✓	✓	✓
• Magazine list		O	✓	✓	✓	✓	✓	✓
• Configurable magazine list		O	✓	✓	✓	✓	✓	✓
• Magazine data		O	✓	✓	✓	✓	✓	✓
• Empty location search and location positioning		O	✓	✓	✓	✓	✓	✓
• Convenient empty location search using softkeys		O	✓	✓	✓	✓	✓	✓
• Loading and unloading of tools		O	✓	✓	✓	✓	✓	✓
• Tool cabinet and tool catalog		–	–	–	–	–	–	–
• Loading and unloading via code carrier system		–	–	–	–	–	–	–
• Adapter data:								
- Turning		✓	✓	✓	✓	✓	✓	✓
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	–	–	✓	✓	✓	✓
• Location-dependent offsets, reference point on wheel:								
- Turning		–	–	–	–	–	–	–
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		✓	✓	✓	✓	✓	✓	✓
• Tool life monitoring and workpiece count		✓	✓	✓	✓	✓	✓	✓
• **Replacement tools for tool management:**	**6FC5800-0AM78-0YH0**							
- Turning		–	O	O	O	O	O	O
- Milling		–	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Multi-tool tool holder		–	–	–	–	–	–	–

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Tool management (continued)								
Manage tools Manage MyTools		–	–	–	–	–	–	–
Identify tool demand	**6FC5800-0AM77-0YH0**							
- Turning		–	O	O	O	O	O	O
- Milling		–	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Tool identification for loading/unloading tools with code carrier								
• **Tool Ident Connection**	**6FC5800-0AP52-0YH0**	–	O	O	O	O	O	O
Communication and data management								
Transfer data to storage medium on rear USB interface of operator panel, e.g. card reader, USB flash drive	Hard disk not possible	✓	✓	–	✓	–	✓	–
Transfer data to storage medium on front USB interface of operator panel, e.g. card reader, USB flash drive	Hard disk not possible	✓	✓	–	✓	–	✓	–
Transfer process data (WRITE ISOPRINT) to USB flash drive		✓	✓	✓	✓	✓	✓	✓
Manage additional drives via:								
• Ethernet 1 Gbps (Windows Share/Linux/FTP)		✓	✓	✓	✓	✓	✓	✓
• USB 3.1 interface		1	2	2	2	2	2	2
• USB 2.0 interface	1 × reserved for machine control panel	2	2	2	2	2	2	2
Peripheral connection via I/O interface based on PROFINET	Only via PP 72/48D PN or PP 72/48D 2/2A PN I/O modules	✓	✓	✓	✓	✓	✓	✓
Connection to an external PROFINET network with SIMATIC PN/PN coupler	**6ES7158-3AD10-0XA0**	O	O	O	O	O	O	O
SIMATIC ET 200SP bus adapter BA 2 × RJ45 2 × RJ45 sockets for PROFINET	**6ES7193-6AR00-0AA0**	–	O	O	O	O	O	O
SIMATIC bus adapter BA LC/FC Media converter glass FOC/copper 1 × LC FO connection 1 × FastConnect (FC) connection for PROFINET	**6ES7193-6AG40-0AA0**	–	O	O	O	O	O	O
Create MyInterface		–	–	–	–	–	–	–
Access MyBackup		–	–	–	–	–	–	–
Production data evaluation:								
• Analyze MyPerformance		–	–	–	–	–	–	–
Host computer connection Server for OPC UA in SINUMERIK Operate Access MyMachine:								
• **Access MyMachine /OPC UA** Variables/ms, maximum number	**6FC5800-0AP67-0YH0**	O 100/200	O 100/200	O 100/200	O 100/200	O 100/200	O 100/200	O 100/200
• **Access MyMachine /OPC UA - Eco**	**6FC5800-0AQ67-0YH0**	O	O	O	O	O	O	O

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Operation

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
Operation								
SINUMERIK operator panels with PCU/TCU		–	–	–	–	–	–	–
SINUMERIK PCU 50.5 Windows 7		–	–	–	–	–	–	–
SIMATIC IPC	See SIMATIC IPC	–	–	✓	–	✓	–	✓
SIMATIC Panels	See SIMATIC Panels	–	–	✓	–	✓	–	✓
Connection for:								
• SIMATIC OPs		–	–	–	–	–	–	–
Control unit management:								
• One operator panel per CNC		✓	✓	✓	✓	✓	✓	✓
• Combinations of several operator panels and several CNCs		–	–	–	–	–	–	–
Handheld units:								
SINUMERIK Handheld Terminal HT 10	**6FC5403-0AA21-0AA1**	–	O	O	O	O	O	O
SINUMERIK Handheld Terminal HT 10 with handwheel	**6FC5403-0AA21-1AA1**	–	O	O	O	O	O	O
• Connection cable for HT 10 - Length 2 m - Length 5 m - Length 8 m - Length 10 m - Length 15 m - Length 20 m - Length 25 m	**6XV1440-4BH20** **6XV1440-4BH50** **6XV1440-4BH80** **6XV1440-4BN10** **6XV1440-4BN15** **6XV1440-4BN20** **6XV1440-4BN25**	–	O	O	O	O	O	O
• Wall holder for HT 10 handheld terminal	**6FC5348-0AA20-0AA0**	–	O	O	O	O	O	O
• Touch pen for HT 10 ELO touch pen system, including adhesive holder	**6AV6881-0AV21-0AA0**	–	O	O	O	O	O	O
• Touch pen caps for ELO touch pen system	**6AV2181-8AV60-0AX0**	–	O	O	O	O	O	O
• Accessories for Handheld Terminal HT 10								
- SIMATIC HMI connection box standard for SINUMERIK HT 10 without emergency stop circuit bridging, wall mounting, PROFINET and PROFIsafe, connection of safety relay, terminal detection, integrated switch, degree of protection IP65	**6AV2125-2AE13-0AX0**	–	O	O	O	O	O	O
- SIMATIC HMI connection box advanced for SINUMERIK HT 10 with emergency stop circuit bridging, wall mounting, PROFINET and PROFIsafe, connection of safety relay, terminal detection, integrated switch, MRP-Client, IRT, degree of protection IP65	**6AV2125-2AE23-0AX0**	–	O	O	O	O	O	O
- SIMATIC HMI connection box distributed for SINUMERIK HT 10 with emergency stop circuit bridging, with Panel Present signal, wall mounting, PROFINET and PROFIsafe, connection of safety relay, terminal detection, integrated switch, MRP-Client, IRT, degree of protection IP65	**6AV2185-5AE23-1AA0**	–	O	O	O	O	O	O
Mini handheld unit with coiled connecting cable	**6FX2007-1AD03**	O	O	O	O	O	O	O
Mini handheld unit with straight cable	**6FX2007-1AD13**	O	O	O	O	O	O	O
• Connection kit for mini handheld unit, non-assembled without Industrial Ethernet	**6FX2006-1BG03**	O	O	O	O	O	O	O
• Connection kit for mini handheld unit, assembled with PROFINET	**6FX2006-1BG20**	O	O	O	O	O	O	O
• 90° angle socket	**6FX2006-1BG56**	O	O	O	O	O	O	O
• Holder for mini handheld unit	**6FX2006-1BG70**	O	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Operation (continued)								
Machine control panels:								
• SINUMERIK MCP 1200U-V	NEW **6FC5303-0AF33-1AA1**	O	O	O	O	O	O	O
• SINUMERIK MCP 398C	**6FC5303-0AF25-0AA0**	–	O	O	O	O	O	O
• SINUMERIK MCP 1500U	NEW **6FC5303-0AF34-1AA1**	O	O	O	O	O	O	O
• SINUMERIK MCP 1200U-H	NEW **6FC5303-0AF32-1AA1**	O	O	O	O	O	O	O
• SINUMERIK MCP Interface PN for customer-specific machine control panel	**6FC5303-0AF03-0AA0**	–	O	O	O	O	O	O
• SINUMERIK MPP Machine Push Button Panel		–	–	–	–	–	–	–
Electronic handwheels:								
• With 120 mm × 120 mm front panel, 5 V DC	**6FC9320-5DB01**	O	O	O	O	O	O	O
• With 76.2 mm × 76.2 mm front panel, 5 V DC	**6FC9320-5DC01**	O	O	O	O	O	O	O
• With 76.2 mm × 76.2 mm front panel, 24 V DC	6FC9320-5DH01	–	–	–	–	–	–	–
• Without front panel, without setting wheel, 5 V DC	**6FC9320-5DF01**	O	O	O	O	O	O	O
• Without front panel, with setting wheel, 5 V DC	**6FC9320-5DM00**	O	O	O	O	O	O	O
• Portable in housing, coiled cable	**6FC9320-5DE02**	O	O	O	O	O	O	O
• Flange socket for portable handwheel	**6FC9341-1AQ**	O	O	O	O	O	O	O
Connection for electronic handwheels to, maximum:		2	3	3	3	3	3	3
• SINUMERIK PPU/PU		2	2	2	2	2	2	2
• SINUMERIK MCP Interface PN	Use: Manual machine	–	1	1	1	1	1	1
Keyboards:								
• Integrated QWERTY keyboard with short-stroke keys		✓	✓	–	✓	–	✓	–
• SINUMERIK keyboards		–	–	–	–	–	–	–
• SINUMERIK ONE KB 1500 full CNC keyboard USB	**6FC5303-0AP47-0AA0**	O	O	O	O	O	O	O
• SINUMERIK ONE KB 1900 full CNC keyboard USB	**6FC5303-0AP47-0BA0**	O	O	O	O	O	O	O
• KBPC CG US standard PC keyboard		–	–	–	–	–	–	–
Connection for external storage devices via USB:								
• USB flash drive 32 GB including SIMATIC IPC-BIOS-Manager	**6AV6881-0AS42-0AA1**	O	O	O	O	O	O	O

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Operation

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Operation (continued)								
Extended operator functions	**6FC5800-0AP16-0YH0**	O	O	O	O	O	O	O
SINUMERIK PCU base software /IPC ≥ V15 for SIMATIC IPC with Microsoft Windows 10 Software option, supply of an electronic license (PDF) Requirement: SIMATIC IPC for SINUMERIK SINUMERIK Operate /PCU (option Q10) or SINUMERIK Operate /PCU Modular (option Q11)	**6FC5800-0AP86-1YH0**	–	–	O	–	O	–	O
SINUMERIK Operate /PC	**6FC5800-0AQ01-0YH0**	–	–	O	–	O	–	O
SINUMERIK Operate /PCU	**6FC5800-0AQ10-0YH0**	–	–	O	–	O	–	O
SINUMERIK Operate /PCU Modular	**6FC5800-0AQ11-0YH0**	–	–	O	–	O	–	O
SINUMERIK Operate Display Manager Multi-window technology for SINUMERIK Operate Single license, without data storage medium Requirement: Resolution > 1920 px × 1080 px	**6FC5800-0AP81-0YH0**	–	–	O	–	O	–	O
Plain text display of user variables		✓	✓	✓	✓	✓	✓	✓
Multi-channel display:								
• Turning		–	–	–	–	–	✓	✓
• Milling		–	–	–	–	–	✓	✓
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	✓	✓

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Operation (continued)								
2D representation of the 3D protection/working areas		✓	✓	✓	✓	✓	✓	✓
Workpiece-related actual value system		✓	✓	✓	✓	✓	✓	✓
Menu selection via the PLC		✓	✓	✓	✓	✓	✓	✓
CNC program messages		✓	✓	✓	✓	✓	✓	✓
Online help for programming, alarms and machine data, expandable		✓	✓	✓	✓	✓	✓	✓
Screen blanking		✓	✓	✓	✓	✓	✓	✓
Access protection		8 levels	8 levels	8 levels	8 levels	8 levels	8 levels	8 levels
Operating software languages:								
• Chinese Simplified, Chinese Traditional, English, French, German, Italian, Korean, Japanese, Portuguese, Spanish		✓	✓	✓	✓	✓	✓	✓
• Additional languages, use of language extensions		✓	✓	✓	✓	✓	✓	✓
• **Language extensions** on DVD-ROM for the operating software SINUMERIK Operate and SINUMERIK 828 e.g. Bulgarian, Croatian, Czech, Danish, Dutch, Finnish, Greek, Hindi, Hungarian, Indonesian, Japanese, Malay, Polish, Romanian, Russian, Slovak, Slovene, Swedish, Tamil, Thai, Turkish, Vietnamese	**On request**	O	O	O	O	O	O	O
SINUMERIK Collection User, manufacturer, and service documentation for download and displayed in the browser, from CNC SW 5.2x Language: English	NEW **6FC5397-0AE15-2BA0**	O	O	O	O	O	O	O
SINUMERIK Collection User, manufacturer, and service documentation for download and displayed in the browser, from CNC SW 5.2x Language: German	NEW **6FC5397-0AE15-2AA0**	O	O	O	O	O	O	O
Monitoring functions								
Working area limitation		✓	✓	✓	✓	✓	✓	✓
Limit switch monitoring Software and hardware limit switches		✓	✓	✓	✓	✓	✓	✓
Position monitoring		✓	✓	✓	✓	✓	✓	✓
Standstill monitoring		✓	✓	✓	✓	✓	✓	✓
Clamping monitoring		✓	✓	✓	✓	✓	✓	✓
2D/3D protection areas		✓	✓	✓	✓	✓	✓	✓
Protect MyMachine /3D Primitives collision avoidance (machine, working area)	**6FC5800-0AS03-0YH0** Only single-channel	–	O	O	O	O	O	O
Protect MyMachine /3D Primitives collision avoidance 1 additional channel	**6FC5800-0AE00-0YH0**	–	–	–	–	–	O	O
Protect MyMachine /3D STL collision avoidance (machine, working area)	6FC5800-0AS02-0YH0	–	–	–	–	–	–	–
Protect MyMachine /Open interface collision avoidance (machine, working area)	6FC5800-0AS04-0YH0	–	–	–	–	–	–	–

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Monitoring functions

Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Monitoring functions (continued)								
Contour monitoring		✓	✓	✓	✓	✓	✓	✓
Contour monitoring with tunnel function		–	–	–	–	–	–	–
Path length evaluation		–	–	–	–	–	–	–
Axis limitation from the PLC		✓	✓	✓	✓	✓	✓	✓
Spindle speed limitation		✓	✓	✓	✓	✓	✓	✓
Extended stop and retract ESR, CNC-controlled and drive-autonomous	**6FC5800-0AM61-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Drive-autonomous extended stop and retract ESR, incl. generator operation	**6FC5800-0AM60-0YH0**	–	O	O	O	O	O	O
IDM integrated tool monitoring and diagnostics		–	–	–	–	–	–	–
DYNEGMA dynamic energy management	**6SL3077-0AA03-0AB0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Compensations								
Backlash compensation		✓	✓	✓	✓	✓	✓	✓
Leadscrew error compensation	Max. 500 interpolation points	✓	✓	✓	✓	✓	✓	✓
Leadscrew error compensation, bidirectional	**6FC5800-0AM54-0YH0** The correctable tolerance band is restricted to 1 mm.	O	O	O	O	O	O	O
Measuring system error compensation		✓	✓	✓	✓	✓	✓	✓
Sag compensation, multi-dimensional	**6FC5800-0AM55-0YH0** The correctable tolerance band is restricted to 1 mm.	–	O	O	O	O	O	O
Quadrant error compensation:								
• Conventional		✓	✓	✓	✓	✓	✓	✓
• With neural networks		✓	✓	✓	✓	✓	✓	✓
Graphic monitoring of the quadrant error compensation using circularity test		✓	✓	✓	✓	✓	✓	✓
Friction compensation with adaptive characteristics	**6FC5800-0AS06-0YH0**	O	O	O	O	O	O	O
Multi Error Compensation Eco	**6FC5800-0AS80-0YH0**	–	O	O	O	O	O	O
Multi Error Compensation Basic	**6FC5800-0AS81-0YH0**	–	O	O	O	O	O	O
Multi Error Compensation Advanced	**6FC5800-0AS82-0YH0**	–	–	–	–	–	O	O
Temperature compensation		✓	✓	✓	✓	✓	✓	✓
Feedforward control, velocity-dependent		✓	✓	✓	✓	✓	✓	✓
Feedforward control, acceleration-dependent:								
• Turning		✓	✓	✓	✓	✓	✓	✓
• Milling		✓	✓	✓	✓	✓	✓	✓
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Backlash compensation, dynamic		✓	✓	✓	✓	✓	✓	✓
Cogging torque compensation for 1 axis/spindle	**6FC5800-0AD50-0YH0**	–	O	O	O	O	O	O
Cylinder error compensation:	Included in option: Technology cycles for grinding Grinding advanced 6FC5800-0AS35-0YH0							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Intelligent dynamic control	**6FC5800-0AS23-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Intelligent load control	**6FC5800-0AS11-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O

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PLC area

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
PLC area								
SIMATIC S7-300		–	–	–	–	–	–	–
SIMATIC S7-200 based (integrated)		✓	✓	✓	✓	✓	✓	✓
Cycle time for PLC		9 ms	9 ms	9 ms	6 ms	6 ms	6 ms	6 ms
Reaction time to process events, terminal to terminal:								
• Turning		7 ms	7 ms	7 ms	7 ms	7 ms	4 ms	4 ms
• Milling		7 ms	7 ms	7 ms	7 ms	7 ms	7 ms	7 ms
• G-Tech Cylindrical		7 ms	7 ms	7 ms	7 ms	7 ms	7 ms	7 ms
• G-Tech Surface		7 ms	7 ms	7 ms	7 ms	7 ms	7 ms	7 ms
Memory expansion Ladder Steps, maximum	Basic configuration: 24000	24000	100000	100000	100000	100000	100000	100000
Memory expansion to 32000 Ladder Steps	**6FC5800-0AD40-0YH0**	–	O	O	O	O	O	O
PLC programming language:								
• Ladder diagram LAD		✓	✓	✓	✓	✓	✓	✓
• Function block diagram FBD		–	–	–	–	–	–	–
• Statement list STL		–	–	–	–	–	–	–
PLC programming tool for integrated PLC	On toolbox DVD-ROM (Article No.: 6FC5830-0CY50-0YA8) or toolbox download version (Article No.: 6FC5830-0EY50-0YA8)	O	O	O	O	O	O	O
PLC Ladder Viewer in SINUMERIK Operate		✓	✓	✓	✓	✓	✓	✓
PLC re-wire Editor in SINUMERIK Operate	INT100/101 only	✓	✓	✓	✓	✓	✓	✓
PLC Ladder Editor in SINUMERIK Operate	PLC program editor	✓	✓	✓	✓	✓	✓	✓
I/O modules:								
• PP 72/48D PN digital I/O module, maximum number:	**6FC5311-0AA00-0AA0**	O	O	O	O	O	O	O
- Turning		1	3	3	4	4	5	5
- Milling		1	3	3	4	4	5	5
- G-Tech Cylindrical/G-Tech Surface		1	3	3	5	5	5	5
• PP 72/48D 2/2A PN digital/analog I/O module, maximum number:	**6FC5311-0AA00-1AA0**	O	O	O	O	O	O	O
- Turning		1	3	3	4	4	5	5
- Milling		1	3	3	4	4	5	5
- G-Tech Cylindrical/G-Tech Surface		1	3	3	5	5	5	5
• General I/Os via PROFIBUS/PROFINET		–	–	–	–	–	–	–
• General SIMATIC PROFINET PLC I/Os		–	–	–	–	–	–	–
• Analog Drive Interface for 4 axes ADI 4	Replaced by retrofit solution: 6FC5300-0BA01-0AA0	–	–	–	–	–	–	–

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
PLC area (continued)								
Digital inputs, maximum		120	216	216	288	288	360	360
Digital outputs, maximum		80	144	144	192	192	240	240
Analog inputs, maximum:								
• Turning		2	6	6	8	8	10	10
• Milling		2	6	6	8	8	10	10
• G-Tech Cylindrical/G-Tech Surface		2	6	6	8	8	10	10
Analog outputs, maximum (including PPU output):								
• Turning		3	7	7	9	9	11	11
• Milling		3	7	7	9	9	11	11
• G-Tech Cylindrical/G-Tech Surface		3	7	7	9	9	11	11
Number of PLC alarms and messages								
• PLC alarms/messages		1248	1248	1248	1248	1248	1248	1248
• Extended PLC alarms/messages		1000	1000	1000	1000	1000	1000	1000
Bit memories, number		4096 (512 bytes)	4096 (512 bytes)	4096 (512 bytes)	4096 (512 bytes)	4096 (512 bytes)	4096 (512 bytes)	4096 (512 bytes)
Timers, number	NEW	512	512	512	512	512	512	512
Counters, number		256	256	256	256	256	256	256
Subroutines		256	256	256	256	256	256	256
FB, FC		–	–	–	–	–	–	–
DB, highest number, max. number	NEW	128	128	128	128	128	128	128
Cyclic block		✓	✓	✓	✓	✓	✓	✓
Cyclic block, servo-synchronous		✓	✓	✓	✓	✓	✓	✓
User machine data for configuring the PLC user program		✓	✓	✓	✓	✓	✓	✓
NCVar selector		✓	✓	✓	✓	✓	✓	✓
Import and export of PLC projects PLC file handling via archives		✓	✓	✓	✓	✓	✓	✓
Freely configurable PLC interface		✓	✓	✓	✓	✓	✓	✓

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Safety functions

Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Safety functions								
Drive-based Safety Integrated Safety functions for personnel and machine protection:								
• Safe Torque Off (STO)		✓	✓	✓	✓	✓	✓	✓
• Safe Brake Control (SBC)		–	✓	✓	✓	✓	✓	✓
• Safe Stop 1 (SS1)		–	✓	✓	✓	✓	✓	✓
Extended Safety Integrated functions: • Safe Torque Off (STO) • Safe Stop 1 (SS1) • Safe Stop 2 (SS2) • Safe Operating Stop (SOS) • Safely-Limited Speed (SLS) • Safe Acceleration Monitor (SAM) / Safe Brake Ramp (SBR) • Safe Speed Monitor (SSM) • Safely-Limited Position (SLP) • Safe Direction (SDI) • Safe Brake Management (SBM) - Safe Brake Control (SBC) - Safe Brake Test (SBT)	**6FC5800-0AC50-0YH0** For one CNC axis/spindle	–	O	O	O	O	O	O
SINAMICS S120 Terminal Module Cabinet TM54F for controlling extended Safety Integrated functions	**6SL3055-0AA00-3BA0** Required for each PPU/PU and NX	–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Commissioning								
Commissioning software for the drive system integrated:								
• SINAMICS S120		–	✓	✓	✓	✓	✓	✓
Auto Servo Tuning AST Fully automatic speed and position controller optimization		✓	✓	✓	✓	✓	✓	✓
SINUMERIK 828D AST addon	**6FC5800-0AS12-0YH0**	–	O	O	O	O	O	O
Commissioning trace integrated Drive optimization without an additional oscilloscope		✓	✓	✓	✓	✓	✓	✓
Standard commissioning via:								
• RS232C serial interface		–	–	–	–	–	–	–
• USB interface with storage medium, e.g. USB flash drive	Hard disk not possible Import/export INI file	✓	✓	✓	✓	✓	✓	✓
• Network drive		✓	✓	✓	✓	✓	✓	✓
• Access MyMachine /P2P for PC/PG	**6FC5860-7YC00-0YA0**	O	O	O	O	O	O	O
STARTER commissioning tool for PC/PG for SINAMICS S120	On toolbox DVD-ROM (Article No.: 6FC5830-0CY50-0YA8) or toolbox download version (Article No.: 6FC5830-0EY50-0YA8)	–	O	O	O	O	O	O
Analyze MyMachine /SINUMERIK Tuning Center								
License Analyze MyMachine /SINUMERIK Tuning Center	**6FC5862-4YC64-0YA0**	–	O	O	O	O	O	O
License Analyze MyMachine /SINUMERIK Expert Trace	**6FC5862-4YC65-0YA0**	–	O	O	O	O	O	O
Analyze MyMachine /SINUMERIK Tuning Center (incl. Expert Trace) Software (without license) Software for path and axis optimization of a machine tool with SINUMERIK 828 including Diagnostic Trace Requirement: SINUMERIK Operate on PC, Operate Commissioning tool (supplied on Toolbox DVD-ROM or download version, Article No. 6FC5830-0CY50-0YA8 or 6FC5830-0EY50-0YA8)	**6FC5862-4YC67-0YA0**	–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Diagnostic functions								
Alarms and messages		✓	✓	✓	✓	✓	✓	✓
Action log can be activated for diagnostic purposes		✓	✓	✓	✓	✓	✓	✓
PLC status		✓	✓	✓	✓	✓	✓	✓
LAD display		✓	✓	✓	✓	✓	✓	✓
PLC remote diagnostics via modem		✓	✓	✓	✓	✓	✓	✓
PLC remote diagnostics via Ethernet		✓	✓	✓	✓	✓	✓	✓
Remote diagnostics and data transfer:								
• **Access MyMachine /P2P RT**	**6FC5800-0AP30-0YH0**	–	O	O	O	O	O	O
	Connection of a modem router to X127							
• **Access MyMachine /P2P** for PC/PG – data transfer between PC/PG and CNCs	**6FC5860-7YC00-0YA0**	O	O	O	O	O	O	O
Remote control of other systems								
Remote desktop control with VNC Viewer		✓	✓	✓	✓	✓	✓	✓
Service and maintenance								
Integrated service planner for the monitoring of service intervals		✓	✓	✓	✓	✓	✓	✓
Easy Extend Simply extends optional machine components		✓	✓	✓	✓	✓	✓	✓
SINUMERIK Ctrl-Energy								
Intelligent standby control of the machine Ctrl-E profiles		✓	✓	✓	✓	✓	✓	✓
Generator operation		✓	✓	✓	✓	✓	✓	✓
Measurement and evaluation of the total energy consumption of the machine and the drive system Ctrl-E analysis:								
• Transfer of manual values from the PLC		✓	✓	✓	✓	✓	✓	✓
• SENTRON PAC2200 measuring device for front panel mounting	**7KM2200-2EA40-1EA1**	O	O	O	O	O	O	O
Ctrl-E Advanced Analysis NEW	**6FC5800-0AQ51-0YH0**	–	O	O	O	O	O	O
Ctrl-E Balanced Spindle Acceleration	**6FC5800-0AQ52-0YH0**	–	O	O	O	O	O	O
Flux reduction:								
• Turning		✓	✓	✓	✓	✓	✓	✓
• Milling		✓	✓	✓	✓	✓	✓	✓
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Reactive-current compensation	Only with SINAMICS S120 Active Line Module	–	–	–	–	–	–	–

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Manufacturer options								
Axis/spindle, each additional	**6FC5800-0AC20-0YH0**	O	O	O	O	O	O	O
Handling channel, each additional	**6FC5800-0AC40-0YH0**	–	O	O	O	O	O	O
Positioning axis/auxiliary spindle, each additional	**6FC5800-0AC30-0YH0**	O	O	O	O	O	O	O
Machining channel, each additional	**6FC5800-0AC10-0YH0**	–	–	–	–	–	O	O
Mode group, each additional:	**6FC5800-0AC00-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
TRANSMIT/cylinder surface transformation	**6FC5800-0AM27-0YH0**	–	O	O	O	O	O	O
TRANSMIT/TRACYL transformation without Y axis	**6FC5800-0AS50-0YH0**	O	O	O	O	O	O	O
Inclined axis:	**6FC5800-0AM28-0YH0**							
• Turning	For non-orthogonal Y axis	–	–	–	O	O	O	O
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical	For non-orthogonal X axis	–	O	O	O	O	O	O
• G-Tech Surface		–	–	–	–	–	–	–
Inclined axis Basic, fixed angle:	**6FC5800-0AS54-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface	For non-orthogonal X axis	–	O	O	O	O	O	O
Oscillation function, non-modal, modal and asynchronous:	**6FC5800-0AM34-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface	CYCLE4071-CYCLE4079	–	O	O	O	O	O	O
Measuring stage 2 Axial measurement, measurements from synchronized actions, cyclic measurement, fast measurement	**6FC5800-0AM32-0YH0**							
• Turning		–	–	–	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	O	O	O	O
Pair of synchronized axes (gantry axes), Basic:	**6FC5800-0AS51-0YH0**	O	O	O	O	O	O	O
• Turning		1	1	1	1	1	1	1
• Milling		1	1	1	1	1	2	2
• G-Tech Cylindrical/G-Tech Surface		1	1	1	1	1	1	1
Travel to fixed stop with Force Control	**6FC5800-0AM01-0YH0**	–	O	O	O	O	O	O
Tangential control:	**6FC5800-0AM06-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Generic coupling CP-Static, e.g. counterspindle:	**6FC5800-0AM75-0YH0**							
• 1 × basic synchronous spindle, coupling ratio 1:1,								
- Turning		–	–	–	–	–	O	O
- Milling		–	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Manufacturer options (continued)								
Generic coupling CP-Basic, e.g. multi-edge turning:	**6FC5800-0AM72-0YH0**							
• 4 axis pairs in simultaneous coupled motion:								
- Turning		O	O	O	O	O	O	O
- Milling		–	–	–	–	–	O	O
- G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
• 1 × synchronous spindle/multi-edge turning:								
- Turning		O	O	O	O	O	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		O	O	O	O	O	O	O
• Master value coupling/curve table interpolation		–	–	–	–	–	–	–
Generic coupling CP-Comfort e.g. electronic gear:	**6FC5800-0AM73-0YH0**							
• 4 axis pairs in simultaneous coupled motion:								
- Turning		–	O	O	O	O	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
• 1 × synchronous spindle/multi-edge turning:								
- Turning		–	O	O	O	O	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
• Electronic gear for 3 leading axes, without curve table, without cascading:								
- Turning		–	O	O	O	O	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
• Axial coupling in the machine coordinate system		–	–	–	–	–	–	–
• Master value coupling/curve table interpolation		–	–	–	–	–	–	–
Leadscrew error compensation, bidirectional	**6FC5800-0AM54-0YH0** The correctable tolerance band is restricted to 1 mm.	O	O	O	O	O	O	O
Sag compensation, multi-dimensional	**6FC5800-0AM55-0YH0** The correctable tolerance band is restricted to 1 mm.	–	O	O	O	O	O	O
Main axis/sub axis Basic for drives	**6FC5800-0AS52-0YH0**	–	O	O	O	O	O	O
• Turning		–	1	1	1	1	2	2
• Milling		–	1	1	1	1	3	3
• G-Tech Cylindrical/G-Tech Surface		–	1	1	1	1	2	2
Evaluation of internal drive variables, Basic	**6FC5800-0AS53-0YH0**	–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available		**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
			SW 24x			**SW 26x**		**SW 28x**	
			PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Manufacturer options (continued)									
SINUMERIK Operate Runtime license OA Easy Screen **Run MyScreens**									
> 5 screens, extended functions		**6FC5800-0AP64-0YH0**							
• Turning			O	O	O	O	O	O	O
• Milling			O	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface			✓	✓	✓	✓	✓	✓	✓
Extended Safety Integrated functions for one CNC axis/spindle		**6FC5800-0AC50-0YH0**	–	O	O	O	O	O	O
Development, commissioning and simulation of virtual machine tools with SINUMERIK 828									
Create MyVirtual Machine /Operate Virtual test environment for SINUMERIK 828	NEW	**MCS4250f** [2] **MCS4250n** [3] **MCS4250a** [4]	O[1]	O[1]	O[1]	O[1]	O[1]	O[1]	O[1]
Create MyVirtual Machine /Open Virtual test environment for SINUMERIK 828 Extension with an open communications interface	NEW	**MCS4255f** [2] **MCS4255n** [3] **MCS4255a** [4]	O[1]	O[1]	O[1]	O[1]	O[1]	O[1]	O[1]
Create MyVirtual Machine /3D Virtual test environment for SINUMERIK 828 Extension comprising a Virtual Machine Creator for JobShop applications	NEW	**MCS4210** [2] **MCS4210n** [3] **MCS4210a** [4]	O[1]	O[1]	O[1]	O[1]	O[1]	O[1]	O[1]
Access protection for cycles Lock MyCycles		**6FC5800-0AP54-0YH0**	O	O	O	O	O	O	O
Extended stop and retract ESR, CNC-controlled and drive-autonomous		**6FC5800-0AM61-0YH0**							
• Turning			–	–	–	–	–	–	–
• Milling			–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface			–	O	O	O	O	O	O
Drive-autonomous extended stop and retract ESR, incl. generator operation		**6FC5800-0AM60-0YH0**	–	O	O	O	O	O	O
Memory expansion to 32000 Ladder Steps		**6FC5800-0AD40-0YH0**	–	O	O	O	O	O	O
Balance cutting:		**6FC5800-0AS05-0YH0**							
• Turning			–	–	–	–	–	O	O
• Milling			–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface			–	–	–	–	–	–	–

[1] In Create MyVirtual Machine, no distinctions are mapped between the hardware types/classes. Projects created with Create/Run MyVirtual Machine can be loaded into SINUMERIK 828 PU/PPU, taking into consideration the HW configurations and constraints.

[2] Floating: Server-based licenses, can be used on multiple work stations (not simultaneously); order recommendation.

[3] Node-locked Server: Server license that is connected to particular clients (work station/PC)
Node-locked Standalone: Local license file that is connected to a particular client (work station/PC)

[4] Academics: For non-commercial use.

For more information, request the detailed description of licenses.

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		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Manufacturer options (continued)								
Friction compensation with adaptive characteristics	**6FC5800-0AS06-0YH0**	O	O	O	O	O	O	O
Multi Error Compensation Eco	**6FC5800-0AS80-0YH0**	–	O	O	O	O	O	O
Multi Error Compensation Basic	**6FC5800-0AS81-0YH0**	–	O	O	O	O	O	O
Multi Error Compensation Advanced	**6FC5800-0AS82-0YH0**	–	–	–	–	–	O	O
CNC lock function	**6FC5800-0AP76-0YH0**	O	O	O	O	O	O	O
DYNEGMA dynamic energy management	**6SL3077-0AA03-0AB0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Advanced Position Control APC ECO	**6FC5800-0AM12-0YH0**	–	O	O	O	O	O	O
Simulated motor encoder	**6FC5800-0AD30-0YH0**	–	O	O	O	O	O	O
Protect MyMachine /3D Primitives collision avoidance (machine, working area)	**6FC5800-0AS03-0YH0** Only single-channel	–	O	O	O	O	O	O
Protect MyMachine /3D Primitives collision avoidance 1 additional channel	**6FC5800-0AE00-0YH0**	–	–	–	–	–	O	O
Protect MyMachine /3D STL collision avoidance (machine, working area)	6FC5800-0AS02-0YH0	–	–	–	–	–	–	–
Protect MyMachine /Open interface collision avoidance (machine, working area)	6FC5800-0AS04-0YH0	–	–	–	–	–	–	–
Nodding compensation ECO • One compensation axis can process one influence quantity • Adaptive compensation via 3 interpolation points	**6FC5800-0AS20-0YH0** No rotary axis possible	–	O	O	O	O	O	O
Nodding compensation ADVANCED • Each compensation axis can process 3 influence quantities • Adaptive compensation via 3 interpolation points • Unlimited number of compensation axes	**6FC5800-0AS21-0YH0** No rotary axis possible	–	O	O	O	O	O	O
Setpoint exchange:	**6FC5800-0AM05-0YH0**							
• Turning		–	–	–	–	–	O	O
• Milling		–	–	–	–	–	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	O	O
Use extended HMI applications Run MyHMI /3GL (maximum 300 MB)	**6FC5800-0AP60-0YH0**	–	O	O	O	O	O	O
Tool identification for loading/unloading tools with code carrier								
• **Tool Ident Connection**	**6FC5800-0AP52-0YH0**	–	O	O	O	O	O	O
Interpolation turning	**6FC5800-0AP57-0YH0**							
• Turning		–	–	–	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Surface turning	**6FC5800-0AR51-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	O	O	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Manufacturer options (continued)								
SMTE cone turning [1] Requirement: CNC software ≥ SW version 4.95	6FC5800-0AR52-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE QR Code [1] Requirement: CNC software ≥ SW version 4.95 SP3	6FC5800-0AR53-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE Engraving [1] Requirement: CNC software ≥ SW version 4.95 SP3	6FC5800-0AR54-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Prime Turning ™	6FC5800-0AR56-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE Chatter avoidance [1] Requirement: CNC software ≥ SW version 4.95 SP4	NEW 6FC5800-0AR57-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE Trochoidal Basic [1] Requirement: CNC software ≥ SW version 4.95 SP4	NEW 6FC5800-0AR58-0YH0							
• Turning		–	–	–	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SMTE Eccentric Basic [1] Requirement: CNC software ≥ SW version 4.95 SP4	NEW 6FC5800-0AR60-0YH0							
• Turning		–	–	–	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	O	O	O	O
SMTE Eccentric Advanced [1] Requirement: CNC software ≥ SW version 4.95 SP4	NEW 6FC5800-0AR61-0YH0							
• Turning		–	–	–	O	O	O	O
• Milling		–	–	–	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	O	O	O	O
Continuous path mode (G64) extended	NEW 6FC5800-0AS37-0YH0							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–

[1] Regarding delivery, please contact your local Siemens sales office or Regional Company.
Requirement:
Execution from external storage EES, Article No. 6FC5800-0AP75-0YH0, or CNC user memory expanded for programs, Article No. 6FC5800-0AP77-0YH0.

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
Manufacturer options (continued)								
Comb grooving	**6FC5800-0AS58-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Intelligent dynamic control	**6FC5800-0AS23-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Intelligent load control	**6FC5800-0AS11-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Jerk adaptation	**6FC5800-0AS22-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SINUMERIK 828D tool orientation (TRAORI, RTCP)	**6FC5800-0AM30-0YH0**							
• Turning		–	–	–	–	–	O	O
• Milling		–	–	–	–	–	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Turning with Y-axis	**6FC5800-0AS56-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Advanced Rapid Movement (COMPPATH) Intelligent adaptation of the machine paths outside of the workpiece	**6FC5800-0AS64-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Reduced dynamic mode	**6FC5800-0AS65-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Run MyRobot /Handling	**6FC5800-0AP72-0YH0**	–	O	O	O	O	O	O
Optimize MyMachining /AdaptiveControl Standard	**6FC5800-0AT00-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Optimize MyMachining /AdaptiveControl Pro	**6FC5800-0AT02-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
Manufacturer options (continued)								
ACM Suite: OMM /ToolMonitor Pro	**6FC5800-0AT03-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
ACM Suite: Multi-spindle extension	**6FC5800-0AT04-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
ACM Suite: Multi-channel extension	**6FC5800-0AT05-0YH0** **MCS2154**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
ACM Suite: Single part extension	**6FC5800-0AT06-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
SINUMERIK PCU base software /IPC ≥ V15 for SIMATIC IPC with Microsoft Windows 10 Software option, supply of an electronic license (PDF) Requirement: SIMATIC IPC for SINUMERIK SINUMERIK Operate /PCU (option Q10) or SINUMERIK Operate /PCU Modular (option Q11)	**6FC5800-0AP86-1YH0**	–	–	O	–	O	–	O
SINUMERIK Operate /PC	**6FC5800-0AQ01-0YH0**	–	–	O	–	O	–	O
SINUMERIK Operate /PCU	**6FC5800-0AQ10-0YH0**	–	–	O	–	O	–	O
SINUMERIK Operate /PCU Modular	**6FC5800-0AQ11-0YH0**	–	–	O	–	O	–	O
Ctrl-E Advanced Analysis NEW	**6FC5800-0AQ51-0YH0**	–	O	O	O	O	O	O
Ctrl-E Balanced Spindle Acceleration	**6FC5800-0AQ52-0YH0**	–	O	O	O	O	O	O
SINUMERIK Operate Display Manager Multi-window technology for SINUMERIK Operate Single license, without data storage medium Requirement: Resolution > 1920 px × 1080 px	**6FC5800-0AP81-0YH0**	–	–	O	–	O	–	O
SINUMERIK 828D AST addon	**6FC5800-0AS12-0YH0**	–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
User options								
Extended functions for grinding Grinding advanced:	**6FC5800-0AS35-0YH0**							
• Dressing (paraxial form-truing) with stock removal cycles (Basic CYCLE95)								
• Cylinder error compensation								
- Turning		–	–	–	–	–	–	–
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Advanced technology functions[1] (expansion of the technology cycles for turning and milling):	**6FC5800-0AP58-0YH0**							
• Pocket milling with free contour definition and islands:								
- Turning		O	O	O	✓	✓	✓	✓
- Milling		O	O	O	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• Stock removal cycles with free contour definition:								
- Turning		O	O	O	✓	✓	✓	✓
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Basic technology functions[3]	NEW **6FC5800-0AQ08-0YH0**	O	✓	✓	✓	✓	✓	✓
Extended operator functions[2]	**6FC5800-0AP16-0YH0**	O	O	O	O	O	O	O
Extended operator basic functions[4]	NEW **6FC5800-0AQ06-0YH0**	O	✓	✓	✓	✓	✓	✓

1) The CNC option Advanced technology functions provides you with technology cycles for the following additional machining operations:
- Asymmetric grooves (turning only)
- Drill and thread milling
- Thread milling
- Multi-edge milling
- Engraving
- Extended stock removal along contour with segmentation of blank (turning only)
- Contour grooving and plunge turning (turning only)
- Milling of contour pockets and spigots with up to 12 islands
- Position pattern - hide position
- Asymmetrically turn a shoulder
- DIN thread undercut

2) The operator functions in the basic scope of the SINUMERIK 828D are designed for standard applications.
The CNC option Extended operator functions enables the following additional operator functions:
- Overstore
- Teach-in
- DRF function
- Extended block search
- Extended skip levels > 2
- Backup of workpiece setup data
- Additional measuring version beyond standard scope (only milling)
  - Standard scope workpiece zero: Set edge, align edge, right-angled corner, 1 hole, 1 circular spigot and rectangular spigot
  - Expansion of the measurement screens via combo box
- Synchronized actions softkey
- MDI load/save

3) The CNC option Basic technology functions provides you with technology cycles of turning and milling for the following additional machining operations:
- Swivel plane and swiveling tool
- Deep hole drilling 2
- Grid or frame position pattern
- Open slot
- Tool management function and magazine list

4) The operator functions in the basic scope of the SINUMERIK 828D are designed for standard applications.
The CNC option Extended operator basic functions enables the following additional operator functions:
- Positioning axes softkey in Jog mode
- Simple face milling/stock remove of the workpiece softkey in Jog mode
- Swiveling softkey in Jog mode
- Manual retraction softkey in Jog mode
- Thread synchronizing softkey in Jog mode
- Active zero offset softkey
- Overview zero offset softkey
- NC/PLC variable softkey
- Bus TCP/IP softkey
- Axis diagnosis sofykey
- Additional measuring version beyond standard scope (only milling)
- Standard scope workpiece zero: Set edge, 1 circular spigot and rectangular spigot

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D						
		SW 24x			SW 26x		SW 28x	
		PPU 26x.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5	PPU 27x.5/ 290.5	PU 272.5
User options (continued)								
Configured stop	**6FC5800-0AS24-0YH0**	–	O	O	O	O	O	O
Program editor:								
• **ShopTurn/ShopMill machining step programming:**	**6FC5800-0AP17-0YH0**							
- Turning		O	O	O	O	O	O	O
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• **programSYNC – multi-channel step sequence programming:**	**6FC5800-0AP05-0YH0**							
- Turning		–	–	–	–	–	O	O
- Milling		–	–	–	–	–	–	–
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Residual material detection and machining for contour pockets and stock removal:	**6FC5800-0AP13-0YH0** Requirement: Advanced technology functions option 6FC5800-0AP58-0YH0							
• Turning		O	O	O	O	O	O	O
• Milling		O	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
3D simulation 1 (finished part):	**6FC5800-0AP25-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Simultaneous recording (real-time simulation of the current machining operation)	**6FC5800-0AP22-0YH0**	O	O	O	O	O	O	O
Measuring cycles for drilling/milling and turning Calibration of workpiece probe, workpiece measurement, tool measurement:	**6FC5800-0AP28-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Operation with tool management:								
• **Replacement tools for tool management:**	**6FC5800-0AM78-0YH0**							
- Turning		–	O	O	O	O	O	O
- Milling		–	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Remote diagnostics and data transfer:								
• **Access MyMachine /P2P RT**	**6FC5800-0AP30-0YH0**	–	O	O	O	O	O	O
Contour handwheel	**6FC5800-0AM08-0YH0**	O	O	O	O	O	O	O
Angle head adapter	**6FC5800-0AM56-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O

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Description ✓ Basic version | O Option | – Not available	Article No. Product ID Notes	SINUMERIK 828D SW 24x PPU 26x.5	SW 24x PPU 27x.5/ 290.5	SW 24x PU 272.5	SW 26x PPU 27x.5/ 290.5	SW 26x PU 272.5	SW 28x PPU 27x.5/ 290.5	SW 28x PU 272.5
User options (continued)								
Identify tool demand	**6FC5800-0AM77-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
SINUMERIK MDynamics:								
• Advanced Surface:	6FC5800-0AS07-0YH0							
- Turning		–	–	–	–	–	–	–
- Milling		✓	✓	✓	✓	✓	✓	✓
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
• **Top Surface:**	**6FC5800-0AS17-0YH0**							
- Turning		–	–	–	–	–	–	–
- Milling		O	O	O	O	O	O	O
- G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Optimized Top Surface [1] NEW	**6FC5800-0AH00-0YH0** Requirement: Top Surface							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Optimized IPO cycle NEW	**6FC5800-0AH01-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	–	–	–	–	–	–
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	O	O
Top Speed plus	**6FC5800-0AS62-0YH0**							
• Turning		–	–	–	–	–	–	–
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Spline interpolation (A, B and C splines)	**6FC5800-0AS16-0YH0**	–	O	O	O	O	O	O
Measure kinematics Determine transformation data of rotary axes:	**6FC5800-0AP18-0YH0**							
• Turning		–	–	–	–	–	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	–	–	–	–	–	–
Host computer connection Server for OPC UA in SINUMERIK Operate Access MyMachine:								
• **Access MyMachine /OPC UA** Variables/ms, maximum number	**6FC5800-0AP67-0YH0**	O 100/200	O 100/200	O 100/200	O 100/200	O 100/200	O 100/200	O 100/200
• **SINUMERIK 828D** **Access MyMachine /OPC UA - Eco**	**6FC5800-0AQ67-0YH0**	O	O	O	O	O	O	O

[1] Not released in combination with Protect MyMachine /3D Primitives (option S03).

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Description ✓ Basic version | O Option | – Not available	**Article No.** **Product ID** Notes	**SINUMERIK 828D**						
		SW 24x			**SW 26x**		**SW 28x**	
		PPU 26x.5	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**	**PPU 27x.5/ 290.5**	**PU 272.5**
User options (continued)								
DXF Reader for PC integrated in SINUMERIK Operate	**6FC5800-0AP56-0YH0**	–	O	O	O	O	O	O
CNC user memory expanded for programs	**6FC5800-0AP77-0YH0**	–	–	–	O 100 MB	O 100 MB	O 100 MB	O 100 MB
Execution from external storage EES	**6FC5800-0AP75-0YH0** Contains the option: CNC user memory expanded for programs 6FC5800-0AP77-0YH0	–	–	–	O	O	O	O
Additional CNC user memory 300 MB, extended Max. extension 1000 MB	**6FC5800-0AD60-0YH0** Requirement: Execution from external storage EES 6FC5800-0AP75-0YH0	–	–	–	O	O	O	O
Monitoring for maximum tool speed/acceleration	**6FC5800-0AS08-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Path acceleration limitation	**6FC5800-0AP26-0YH0**							
• Turning		–	O	O	O	O	O	O
• Milling		–	O	O	O	O	O	O
• G-Tech Cylindrical/G-Tech Surface		–	O	O	O	O	O	O
Simulation of virtual machine tools with SINUMERIK 828 during production								
Run MyVirtual Machine /Operate NEW Control-identical programming station for NC programs or for training and apprenticeship purposes	**MCS4220** [1] **MCS4220n** [2] **MCS4220a** [3]	O	O	O	O	O	O	O
Run MyVirtual Machine /Open NEW Extension with an open communications interface for integration into a simulation system	**MCS4230** [1] **MCS4230n** [2] **MCS4230a** [3]	O	O	O	O	O	O	O
Run MyVirtual Machine /3D NEW Extension to the virtual machine during work preparation	**MCS4240** [1] **MCS4240n** [2] **MCS4240a** [3]	O	O	O	O	O	O	O

[1] Floating: Server-based licenses, can be used on multiple workstations (not simultaneously); order recommendation.

[2] Node-locked Server: Server license that is connected to particular clients (work station/PC)
Node-locked Standalone: Local license file that is connected to a particular client (work station/PC)

[3] Academics: For non-commercial use.

For more information, request the detailed description of licenses.

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Product name	SINUMERIK 828D			
	SW 24x		SW 26x	SW 28x
	PPU 26x.5	PPU 27x.5/ PPU 290.5/ PU 272.5	PPU 27x.5/ PPU 290.5/ PU 272.5	PPU 27x.5/ PPU 290.5/ PU 272.5
Number of DRIVE-CLiQ ports	–	3	3	3
Number of PROFINET ports (PLC periphery interface)	1	2	2	2
Maximum axes in system (including axes/spindles in machining channel + axes in handling channel + PLC positioning axes)				
• Turning	5	8	11	18[2]
• Milling	5	8	11	16[1]
• G-Tech Cylindrical/G-Tech Surface	5	8	11	18[2]
Number of axes/spindles (basic scope)				
• Turning	3	3	3	3
• Milling	4	4	4	4
• G-Tech Cylindrical/G-Tech Surface	3	3	3	3
Number of axes with drive-based Safety Integrated, maximum (Extended Safety Integrated functions)				
• Turning	5	5	6	6 / 12[2]
• Milling	5	5	6	6 / 12[1]
• G-Tech Cylindrical/G-Tech Surface	5	5	6	6 / 12[2]
Maximum number of axes/spindles + PLC positioning axis in machining channel(s)				
• Turning	5	5	6 + 2	10 + 2
• Milling	5	5	6 + 2	8 + 2
• G-Tech Cylindrical/G-Tech Surface	5	5	6 + 2	10 + 2
Maximum number of axes in handling channel(s)				
• Turning	–	4	4	4 + 4
• Milling	–	4	4	4 + 4
• G-Tech Cylindrical/G-Tech Surface	–	4	4	4 + 4
IPO cycle for max. configuration				
• Turning	9 ms	9 ms	6 ms	6 ms
• Milling	9 ms	9 ms	6 ms	3 ms
• G-Tech Cylindrical/G-Tech Surface	9 ms	9 ms	6 ms	3 ms
Minimum block change time, approx.				
• Turning	9 ms	9 ms	6 ms	6 ms
• Milling	9 ms	9 ms	6 ms	3 ms
• G-Tech Cylindrical/G-Tech Surface	9 ms	9 ms	6 ms	3 ms
Minimum block change time with compressor, approx.				
• Turning	9 ms	9 ms	6 ms	6 ms
• Milling (depending on the part program)	3 ms	0.9 ms	0.6 ms	0.3 ms
• G-Tech Cylindrical/G-Tech Surface		–	–	–
Position control cycle[3]				
• Turning	3 ms (125 µs)	3 ms (125 µs)	3 ms (125 µs)	1.5 ms (125 µs)
• Milling	3 ms (125 µs)	3 ms (125 µs)	3 ms (125 µs)	3 ms (125 µs)
• G-Tech Cylindrical/G-Tech Surface	3 ms (125 µs)	3 ms (125 µs)	3 ms (125 µs)	3 ms (125 µs)
PLC cycle time	9 ms	9 ms	6 ms	6 ms
DRIVE-CLiQ S120: Speed and current controller clock cycle	–	125 µs	125 µs	125 µs
Speed and current controller clock cycle for a high-speed spindle				
Mixed operation without NX: 4 × 125 µs and 1 × 62.5 µs, max. number of axes = 5				
• Turning	–	–	–	–
• Milling	–	62.5 µs	62.5 µs	62.5 µs
• G-Tech Cylindrical/G-Tech Surface	–	62.5 µs	62.5 µs	62.5 µs
SINAMICS S200: Speed and current controller clock cycle for a high-speed spindle	125 µs	–	–	–

1) The maximum number of axes/spindles can be increased to 10; 6 axes/spindles can be connected to the PPU 270.5/271.5 or PPU 290.5 or PU 272.5 and 4 axes/spindles to the NX15.3.

2) The maximum number of axes/spindles can be increased to 12; 6 axes/spindles can be connected to the PPU 270.5/271.5 or PPU 290.5 or PU 272.5 and 6 axes/spindles to the NX15.3.

3) With active Dynamic Servo Control (DSC), the position control cycle corresponds to the current controller cycle of 125 µs.

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Non-Volatile Random-Access Memory (NVRAM) for:				
• OEM	512 KB	512 KB	512 KB	512 KB
• User data	3 MB	3 MB	5 MB	10 MB
Number of Numeric Control Extensions NX10.3				
• Turning	–	1[4)]	2	2
• Milling	–	1[4)]	2	2
• G-Tech Cylindrical/G-Tech Surface	–	1[4)]	2	2
Number of Numeric Control Extensions NX15.3				
• Turning	–	1[4)]	1	2
• Milling	–	1[4)]	1	2
• G-Tech Cylindrical/G-Tech Surface	–	1[4)]	1	2
Number of I/O modules PP 72/48D PN or PP 72/48D 2/2A PN				
• Turning	1	3	4	5
• Milling	1	3	4	5
• G-Tech Cylindrical/G-Tech Surface	1	3	5	5

[4)] Only for handling channel.

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Notes